A
Project Report
on

Pharma and Counterfeit Drug Prevention Supply Chain and Smart Contracts System in Blockchain

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ABSTRACT

In the healthcare industry, the integrity of pharmaceutical supply chains is paramount to ensure the safety and efficacy of drugs. However, the proliferation of counterfeit drugs poses significant risks to public health and undermines trust in medical treatments. This project introduces a revolutionary approach to tackling these challenges through the implementation of a Blockchain-based Supply Chain Management System integrated with Smart Contracts. The proposed system leverages Blockchain technology to create an immutable, decentralized ledger for tracking pharmaceutical products throughout their entire lifecycle—from production to end-user. By recording every transaction and movement on this transparent ledger, the system enhances traceability and accountability, significantly reducing the risk of counterfeit drugs entering the market. Smart Contracts are employed to automate and enforce compliance with regulatory standards and contractual agreements. These self-executing contracts ensure that all participants in the supply chain—manufacturers, distributors, retailers, and regulatory authorities—adhere to predefined rules and standards without manual intervention. This not only speeds up processes but also minimizes human error and fraud. The system provides real-time visibility into the status and provenance of each drug, empowering stakeholders with accurate and timely information. End-users can verify the authenticity of their medications by scanning product codes, thereby safeguarding against counterfeit products. Furthermore, the system's secure data sharing capabilities protect sensitive information while facilitating seamless collaboration among all parties involved. By integrating Blockchain technology and Smart Contracts, this project aims to establish a more secure, efficient, and transparent pharmaceutical supply chain. It addresses critical issues related to counterfeit drugs and regulatory compliance, ultimately contributing to improved patient safety and enhanced trust in pharmaceutical products.

Keywords: Blockchain Technology, Smart Contracts, Pharmaceutical Supply Chain, Counterfeit Drug Prevention, Healthcare Security, Data Integrity, Drug Authentication, Supply Chain Transparency, Real-Time Tracking, Secure Data Sharing, Fraud Prevention, Traceability, etc.

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ABBREVIATIONS

Abbreviation	Illustration
OTP	One Time Password
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
HTML	HyperText Markup Language
2FA	Two-Factor Authentication

Chapter 1

Introduction

1.1 Overview

The pharmaceutical industry faces critical challenges related to the integrity and safety of its supply chain, notably the threat of counterfeit drugs. These counterfeit products not only endanger public health but also undermine trust in the pharmaceutical system and contribute to significant economic losses. Traditional supply chain management systems often struggle with issues of transparency, traceability, and compliance, which exacerbate the risk of counterfeit drugs and inefficient regulatory enforcement. To address these issues, this project proposes a cutting-edge solution leveraging Blockchain technology combined with Smart Contracts. Blockchain provides a decentralized, immutable ledger that records every transaction and movement of pharmaceutical products, ensuring transparency and end-to-end traceability. This technology allows stakeholders to track drugs from production through distribution to the end-user, significantly reducing the risk of counterfeit products entering the market. Meanwhile, Smart Contracts automate and enforce compliance with regulatory standards, streamline processes, and prevent fraud by executing predefined rules without manual intervention. Together, these technologies promise to enhance the security, efficiency, and reliability of the pharmaceutical supply chain, ultimately protecting public health and strengthening confidence in medical products.

1.2 Overview of Blockchain

Blockchain is a system of recording information in a way that makes it difficult or impossible to change, hack, or cheat the system.
A blockchain is essentially a digital ledger of transactions that is duplicated and distributed across the entire network of computer systems on the blockchain. Each block in the chain contains a number of transactions, and every time a new transaction occurs on the blockchain, a record of that transaction is added to every participant's ledger. The decentralised database managed by multiple participants is known as Distributed Ledger Technology (DLT).
Blockchain is a type of DLT in which transactions are recorded with an immutable cryptographic signature called a hash.

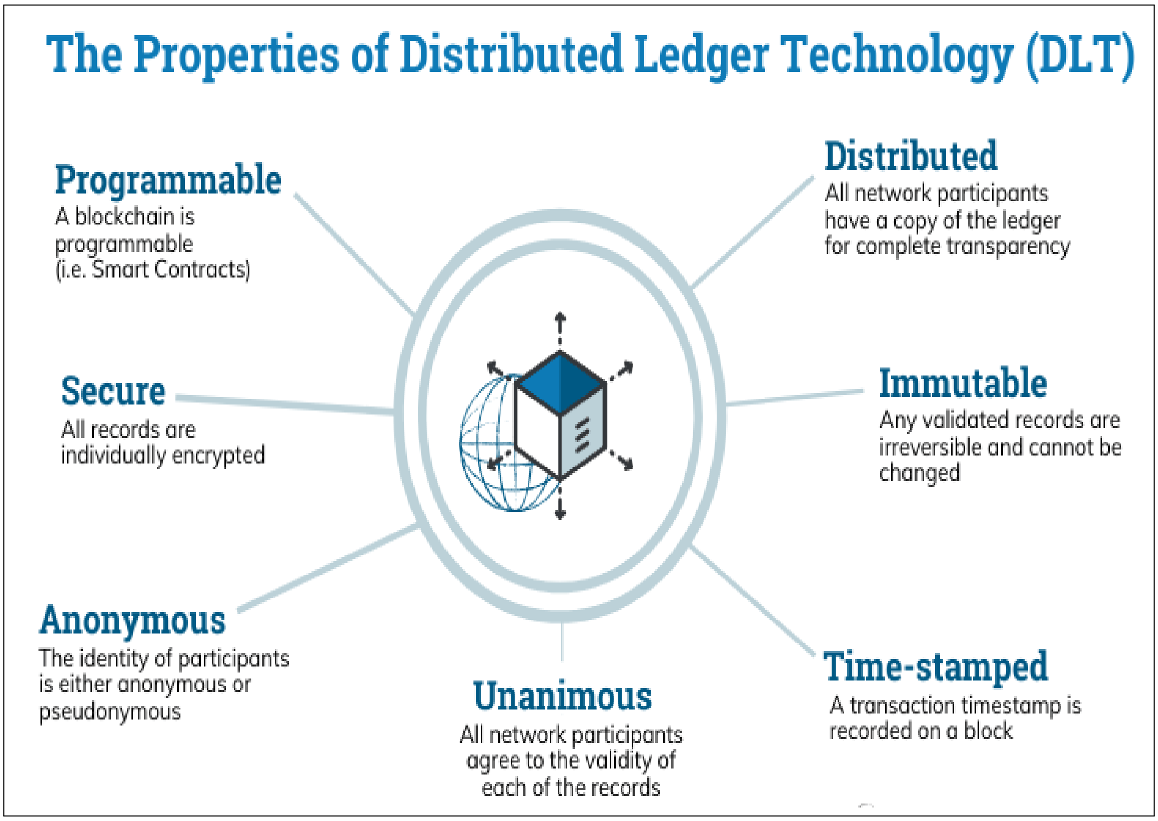

Figure 1.1: Blockchain Technology (DLT)

This means if one block in one chain was changed, it would be immediately apparent it had been tampered with. If hackers wanted to corrupt a blockchain system, they would have to change every block in the chain, across all of the distributed versions of the chain.

Blockchains such as Bitcoin and Ethereum are constantly and continually growing as blocks are being added to the chain, which significantly adds to the security of the ledger.

1.2.1 What is Blockchain Technology?

In a few words, a blockchain is a digital ever-growing list of data records. Such a list is comprised of many blocks of data, which are organized in chronological order and are linked and secured by cryptographic proofs.

The first prototype of a blockchain is dated back to the early 1990s when computer scientist Stuart Haber and physicist W. Scott Stornetta applied cryptographic techniques in a chain of blocks as a way to secure digital documents from data tampering. The work of Haber and Stornetta certainly inspired the work of Dave Bayer, Hal Finney, and many other computer scientists and cryptography enthusiasts - which eventually lead to the creation of Bitcoin, as the first decentralized electronic cash system (or simply the first cryptocurrency). The Bitcoin whitepaper was published in 2008 under the pseudonym Satoshi Nakamoto.

Although the blockchain technology is older than Bitcoin, it is a core underlying component of most cryptocurrency networks, acting as a decentralized, distributed and public digital ledger that is responsible for keeping a permanent record (chain of blocks) of all previously confirmed transactions.

Blockchain transactions occur within a peer-to-peer network of globally distributed computers (nodes). Each node maintains a copy of the blockchain and contributes to the functioning and security of the network. This is what makes Bitcoin a decentralized digital currency that is borderless, censorship-resistant, and that does not require third-party intermediation.

As a distributed ledger technology (DLT) the blockchain is intentionally designed to be highly resistant to modification and frauds (such as double-spending). This is true because the Bitcoin blockchain, as a database of records, cannot be altered, nor can it be tampered without an impractical amount of electricity and computational power - which means the network can enforce the concept of "original" digital documents, making each Bitcoin a very unique and un-copyable form of digital currency.

The so-called Proof of Work consensus algorithm is what made it possible for Bitcoin to be built as a Byzantine fault tolerance (BFT) system, meaning that its blockchain is able to operate continuously as a distributed network, even if some of the participants (nodes) present dishonest behavior or inefficient functionality. The Proof of Work consensus algorithm is an essential element of the Bitcoin mining process.

The technology of blockchain may also be adapted and implemented in other activities, such as healthcare, insurance, supply chain, IOT, and so on. Although it was designed to operate as a distributed ledger (on decentralized systems), it may also be deployed on centralized systems as a way to assure data integrity or to reduce operational costs.

1.2.2 Why Blockchain Technology?

Comparative Study In traditional storage where all data is store in only one place called centralized storage, the chances of data security are less as compared to decentralized storage due to the owner of centralized storage could monitor the data and it can be altered or theft. With the quick requirement to access data, each individual wants to retrieve their data instantly and more securely for this reason decentralized storage is developed. In decentralized storage data is stored in more than one data block; this itself reduces the chances of data reaching the data stealers. Because data stealers unaware of where the rest of the data is stored, due to this reason decentralized storage is popular and used widely.

As shown in Fig.1.3 all data user who wants to access their data is dependent on only one server. Due to this bottleneck can arises and results in a longer time needed to access data because of high load on single point server.

There is also limited numbers of the user who can connect the server at the same time, the application where many numbers of user's needs to connects and access server are not possible, and if this server caught by some hardware failure or by some

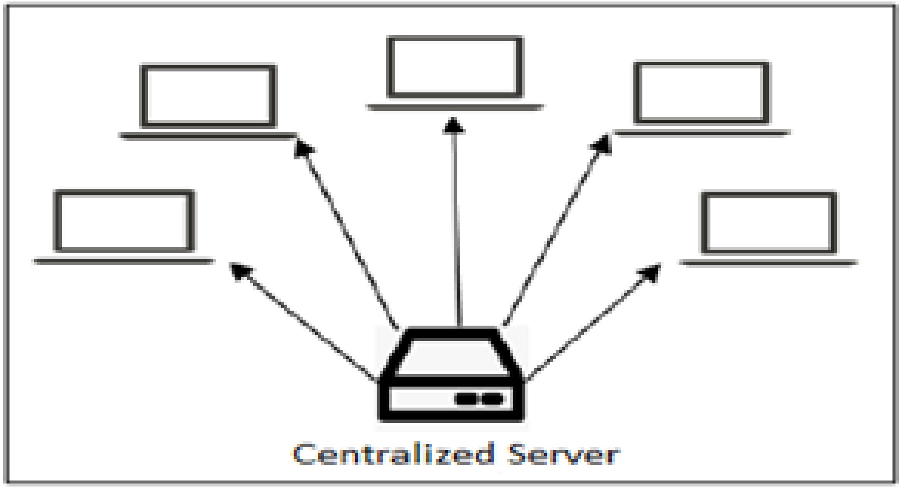

Figure 1.2: Traditional Centralized Storage

fault in setup, the data stored on the server will be inaccessible. If this kind of situation occurs and data may unexpectedly lose then it is very difficult to retrieve data back. The maintenance of this server is also high. Another issue with this data storing technique is that they charge the highest possible price to their customer to use their services. Due to overcome all the situations faced in centralized storage, decentralized storage is used to remove all the difficulty which is faced in a centralized structure.

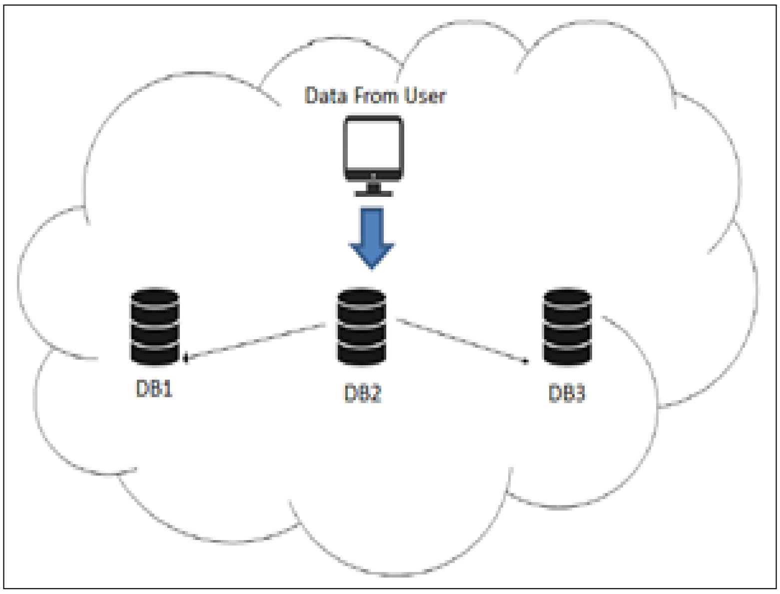

Figure 1.3: Decentralized Storage

The structure of decentralized storage is shown in Fig.3. Data from the user is being stored on multiple databases assign by a server to make a replica of data, while storing data on a different server it is broken into a piece of chunks. This means, there is huge storage space to store data, and the cost of storing data on a server is less. There is no single party to own the data and control it. This method of storing data is helpful when hackers want to steal data stored on cloud storage. Because of this, they won't able to retrieve complete data. As data stored data at a different location is itself secure in the senses of data loss but to protect data from hackers, we need an encryption method to make data more secure. Even a single data file is stored at multiple locations with small data blocks, but what if that small block of the file contains sensitive information. For this reason, we need an encryption method to encrypt data. All these storage uses some sort of algorithm to make data more secure for their customers. Even a hacker able to access a piece of data, due to an encryption mechanism used by the service provider's hackers is unable to decrypt the information within the data block. That's why decentralized storage is preferred as compared to a centralized storage structure. [1]

1.2.3 Use of Blockchain Technology

A blockchain system may be considered as a simply incorruptible cryptographic database where vital and confidential user's information will be recorded. The system is maintained by a network of computers, which is accessible to anyone running the software. Blockchain operates as a pseudo-anonymous system that has nonetheless privacy problem in view that all transactions are exposed to the general public, even though it is tamper-proof inside the sense of data-integrity. The access control to manage heterogeneous user's confidential records across a couple of MNC establishments and devices had to be cautiously designed. Blockchain itself isn't designed as a massive-scale storage system. Within the context of framework for secure banking, a decentralized storage solution would significantly complement the weak point of blockchain within the perspective. The blockchain network as a decentralized system is extra resilient in that there is no single-point assault or failure compared to centralized systems. However, because all the bitcoin transactions are public and everyone has got right of entry to, there already exists analytics equipment that picks out the members within the community based totally on the transaction records [2].

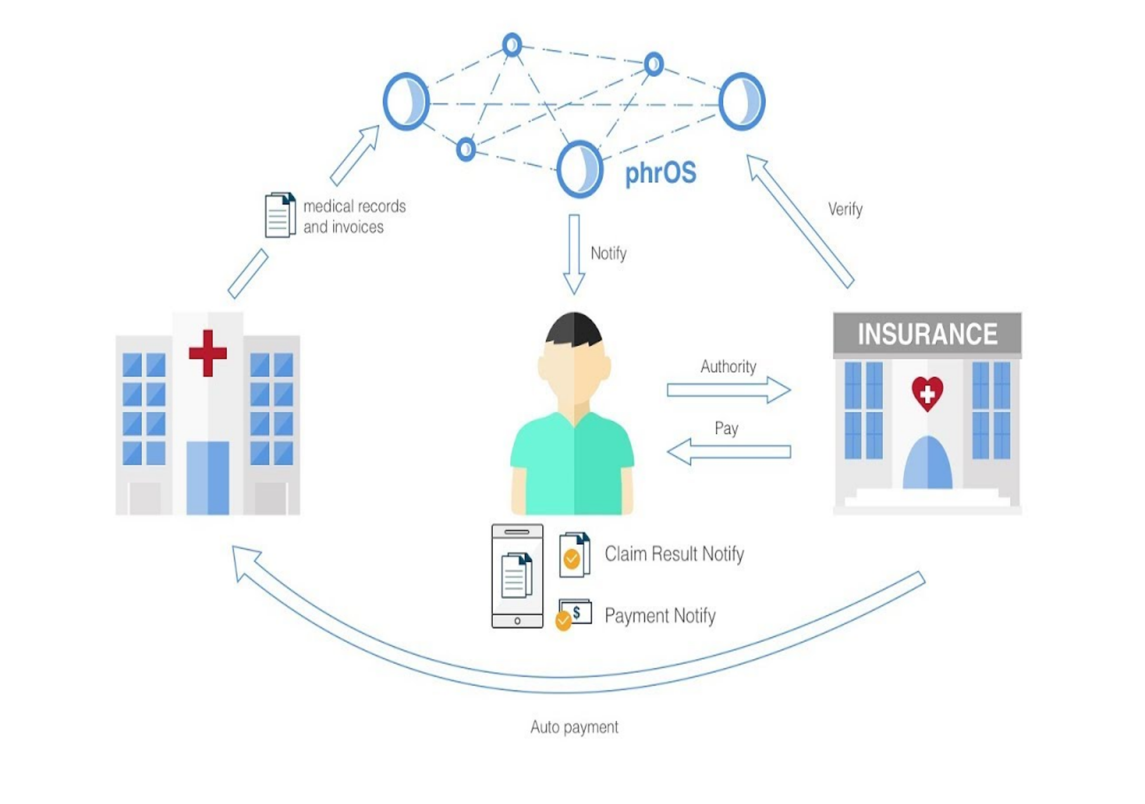

Figure 1.4: Blockchain In Healthcare

In this research work fig.1.5 shows the system overview, and the most important module is blockchain implementation comprises two kinds of records: blocks and transactions. In every block contains a timestamp and a link to a preceding block is supplied via the secure hash algorithm. During the storage, the transaction information into the blockchain system executes various algorithms like SHA for hash generation, mining for generating a valid hash, smart contract for system policy, and consensus for validating current blockchain on all Peer to Peer nodes. Therefore, banking application is more secure. Second thing is that data storage and accessibility. For this point use the Secret Shamir hashing technique and keyword as well as content-based cryptography techniques.

1.2.4 Need of Blockchain in Security

The major question that arises is to why use blockchain when already the market has flourishing plethora of other databases. What substantial importance it holds against the competing products. For this let's understand the problem with the existing systems. They could be summed up as follows:

- Difficult to monitor and evaluate asset ownership and its transfer in a trusted business network.
- Inefficient, expensive, vulnerable: All these factors extremely hinder the performance and there by destroying the progress.

The below issues basically we facing in banking data security and transaction approach which defined here

- To migrate the centralization of banking transaction into the decentralized approach.
- To create a single platform where user can access all bank accounts using transactional authentication.
- To eliminate all physical things dependency which is must require for banking transaction.
- To implement such approach on global environment using secure time less time consuming manner.
- We notice that the decentralized architecture provides the automatic data recovery from different attacks.

To create the blockchain based un-modifiable certificates, initially the university needs to get registered. Any transaction can be sent through the wallet address of the registered university. Only the owner of the smart contract has the authority to add the universities. Once added the university, will be able to access the system and can create certificates with data fields. Each created certificate will be stored in the Inter planetary file system (IPFS). It will then return the unique hash generated using SHA-256 algorithm. This will serve as unique identity for each document. This generated hash and detail of certificates will be stored in the blockchain and the student will be provided with the resultant transaction id. Anyone can use this transaction id to verify the certificate details and can view the original copy of certificate using IPFS hash stored along with data. And it is not possible to modify this certificate or to create fake certificate using the same data. Hence with this we can solve the problem of certificate forgery.

1.3 Motivation

The motivation for this project arises from the urgent need to combat counterfeit drugs, which pose significant health risks and undermine trust in the pharmaceutical industry. Current supply chain systems often lack the transparency and efficiency required to effectively track and verify drug authenticity. By leveraging Blockchain technology and Smart Contracts, this project aims to create a secure, transparent, and automated system that enhances traceability, reduces fraud, and ensures compliance. This innovative approach seeks to safeguard public health and improve confidence in pharmaceutical products by addressing the critical vulnerabilities in existing supply chain practices.

1.4 Problem statement

To develop Pharma – counterfeit drugs Prevention Supply Chain System.

1.5 Objectives

The objectives of the system are

- **To develop** a custom Blockchain solution tailored for the healthcare pharmaceutical supply chain to ensure real-time tracking and verification of drugs.
- **To integrate** Smart Contracts into the custom Blockchain system to automate compliance with healthcare regulations and reduce manual processing errors.
- **To enhance** supply chain transparency and accountability by using an immutable Blockchain ledger to record every transaction involving pharmaceutical products.
- **To implement** advanced counterfeit detection mechanisms within the Blockchain framework to prevent the introduction of fake drugs into the healthcare system.
- **To enable** secure and timely access to accurate drug information for all stakeholders, thereby improving trust and operational efficiency in the pharmaceutical supply chain.

1.6 Organization of report

This topic describes general overview about the residential management system, application, necessary and objective.

Literature Survey

This chapter describe about survey of all the literature based on many existing system have been studied which provide some drawbacks of their methods and algorithms.

Software Requirement Specification

This chapter describe problem description in detail. It contains the hardware and software requirements of the project and also contain UML diagram that more elaborate the system.

Design of Software

This chapter introduces architecture of the system and module of the system. It also contain the registration DFD and UML diagram are explained in this chapter.

Prototype model of project

This chapter describe prototype model of system. This chapter discuss about how the system is implemented and a general overview of system working implemented.

Conclusion

This chapter concludes the project.

Chapter 2

Literature Survey

This chapter discuss brief literature regarding the project. Literature survey is mainly used to identify information relevant to the project work and know impact of it within the project area. It defines as till yet how many surveys have been done knowledge of latest technology and implementation designs.

2.1 Literature Survey Table

Table 2.1: Literature Survey Table

Sr.No	Title	Year	Author	Details
1	Blockchain-based solutions for the pharmaceutical supply chain: A review	2019	Wang, Y., Han, J., & Bao, H.	This system gives patients a comprehensive, immutable log and easy access to their medical information across providers and treatment sites.
2	Applying blockchain technology to healthcare and pharmaceutical supply chains	2019	Armer, J. M., & Catterall, S.	The technology is more scalable, tamper proof and time stamped, making health data more secure.

- **Bauwens, L., & Bruynseels, K. (2020):** This paper provides a comprehensive review of existing literature on blockchain applications in the pharmaceutical industry. The authors highlight the technology's potential to enhance transparency, traceability, and security in the supply chain. They identify key challenges, such as scalability and regulatory compliance, while emphasizing the need for collaboration among stakeholders to fully leverage blockchain's benefits in combating counterfeit drugs.
- **Armer, J. M., & Catterall, S. (2019):** In this study, the authors explore practical applications of blockchain technology specifically within healthcare and

pharmaceutical supply chains. They discuss how blockchain can facilitate secure transactions, improve inventory management, and enhance the traceability of products. The paper outlines various use cases and highlights the importance of integrating blockchain with existing systems to maximize its effectiveness in preventing counterfeit drugs.

- **Gordon, W. J., & Catalini, C. (2018):** This systematic review examines the role of blockchain technology in healthcare, focusing on its implications for patient data management and supply chain integrity. The authors analyze various studies to assess the benefits and limitations of blockchain in healthcare settings. They identify significant opportunities for improving trust and data sharing while noting challenges related to implementation and standardization that must be addressed to ensure success.

- **Deloitte (2020):** This report from Deloitte discusses the future of blockchain in healthcare, particularly its potential to enhance transparency and efficiency in the pharmaceutical supply chain. It highlights specific use cases, including tracking drug authenticity and managing inventory. The report emphasizes that by adopting blockchain, organizations can significantly reduce the risks associated with counterfeit medications, thereby improving patient safety and operational efficiency.

- **Zheng, Z., Zheng, Y., & Wu, J. (2020):** This survey provides a broad overview of blockchain applications in healthcare, focusing on its opportunities and threats. The authors categorize various applications, including drug traceability and patient data management, while also discussing potential challenges such as privacy concerns and technical limitations. The paper serves as a valuable resource for understanding the multifaceted impact of blockchain on healthcare, particularly in addressing the issues surrounding counterfeit drugs.

- **Zheng et al. (2018)** highlighted that blockchain technology offers a decentralized and immutable ledger that can be instrumental in securing supply chains. Their study emphasized that traditional pharma supply chains lack transparency, which gives counterfeiters an easy entry point. By integrating blockchain, each drug movement can be verified at every node, preventing unauthorized tampering or duplication of medicines.

- **Tseng et al. (2018)** explored how smart contracts on Ethereum can automate supply chain operations in pharmaceutical logistics. They concluded that smart contracts can validate actions like shipping, receiving, and payments without the need for human intervention. This automation minimizes delays, enhances trust between parties, and ensures compliance with regulations in real-time.

- **Ramya and Varalakshmi (2020)** proposed a blockchain-based drug verification system where each medicine pack is tagged with a unique digital ID. Their system used QR codes linked to blockchain entries, which consumers could scan to confirm the authenticity of the drug. Their implementation showed a significant improvement in consumer trust and detection of counterfeit drugs in retail outlets.

- **Patra et al. (2021)** discussed the challenges faced by regulatory bodies like the FDA in tracking counterfeit drugs in global markets. Their work showed that blockchain can provide end-to-end visibility of drug origin, storage conditions, and transport data. This transparency makes it easier for regulators and manufacturers to conduct audits and identify weak links in the distribution chain.

- **Rathi and Malviya (2022)** reviewed existing blockchain implementations in industries and proposed a Hyperledger-based framework for pharma supply chains. Their study found that permissioned blockchains are better suited for pharmaceutical applications due to the need for privacy, scalability, and secure access controls among trusted parties in the ecosystem.

2.2 Summary of Literature

The project addresses critical existing problems associated with counterfeit drugs in the pharmaceutical industry. Counterfeit medications pose significant risks to public health, leading to ineffective treatment, adverse reactions, and even fatalities. The prevalence of fake drugs undermines trust in healthcare systems and increases the burden on regulatory authorities. Moreover, the complex and often opaque supply chain makes it challenging to trace the origins and authenticity of pharmaceuticals, allowing counterfeit products to infiltrate legitimate markets. By implementing a blockchain-based system with smart contracts, this project aims to enhance transparency, accountability, and traceability in the drug supply chain, ultimately protecting consumers and ensuring access to safe and effective medications.

The existing systems for managing the pharmaceutical supply chain primarily rely on traditional methods that often lack transparency and efficiency. Many pharmaceutical companies use centralized databases to track drug production and distribution, but these systems are vulnerable to manipulation and data breaches. Current verification processes for ensuring drug authenticity typically involve manual checks, which can be time-consuming and error-prone. Additionally, the complexity of the supply chain, with multiple stakeholders involved—from manufacturers to distributors and pharmacies—makes it difficult to trace the origins of medications effectively.

- **Centralized Databases:** Most pharmaceutical supply chains rely on centralized systems to store and manage data, which can be prone to data breaches and manipulation.

- **Manual Verification Processes:** Current methods for verifying drug authenticity often involve manual checks, leading to inefficiencies and higher chances of human error.

- **Limited Traceability:** Tracking the movement of pharmaceuticals through the supply chain is challenging, making it difficult to trace counterfeit drugs back to their source.

- **Lack of Transparency:** The complexity of the supply chain, involving multiple stakeholders (manufacturers, wholesalers, distributors, and pharmacies), results in a lack of visibility into each step of the process.
- **High Costs of Counterfeit Drugs:** The presence of counterfeit medications leads to increased healthcare costs due to ineffective treatments and additional patient harm, straining healthcare systems.
- **Limited Consumer Awareness:** Many consumers lack the tools or knowledge to verify the authenticity of their medications, increasing their vulnerability to counterfeit products.
- **Slow Response to Fraud:** The current systems struggle to respond quickly to instances of fraud or counterfeit detection, delaying necessary interventions.

These limitations underscore the need for a more secure, transparent, and efficient system to prevent counterfeit drugs in the pharmaceutical supply chain.

The literature extensively supports the integration of blockchain technology into pharmaceutical supply chains as a promising solution to combat counterfeit drugs. Multiple studies have revealed that traditional systems suffer from a lack of transparency, traceability, and real-time verification, allowing counterfeiters to infiltrate the supply chain. Researchers argue that blockchain's decentralized and immutable nature ensures that each transaction or movement of drugs is permanently recorded, making it difficult to alter records or inject fake medicines into the system.

Several papers have focused on the use of smart contracts to automate key logistics operations such as verification, shipment confirmation, and compliance enforcement. Smart contracts, built on platforms like Ethereum, can trigger predefined actions—such as payment release or alert generation—when specific conditions are met. This reduces human error, speeds up processing, and improves trust between manufacturers, distributors, and pharmacies.

Another key area explored in recent works is consumer verification. Techniques like QR-code-based systems, where consumers scan a code linked to a blockchain record, have proven effective. This allows end-users to instantly verify the origin, batch number, and authenticity of a drug. Studies also highlight the use of IoT integration (like temperature sensors) to track storage conditions of sensitive medicines and record that data immutably on the blockchain.

Overall, the literature provides a strong foundation for developing a blockchain-based pharma system that not only prevents counterfeit drugs but also improves supply chain efficiency. The combination of traceability, automation through smart contracts, and end-user verification significantly enhances the integrity of the pharmaceutical ecosystem. These findings validate the need and feasibility of the proposed system, offering real-world potential for improved drug safety and trust in global healthcare delivery.

Chapter 3

Software Requirement Specification

This chapter is about the software requirement specification.It contain the overall problem Statement of the project and the hardware requirements, software requirements and functional and non-functional requirements of the project.

3.1 Software Specification

3.2 Software Requirements

- Operating system: Windows XP/7 Higher
- Programming Language: Java 1.8
- Tools: Eclipse Oxygen, XAMPP, Tomcat 8.5
- Database: MySQL 5.1

3.3 Hardware Requirements

- System : Windows 11.
- Hard Disk : 240 GB.
- Monitor : 15 VGA Colour.
- IO Devices : Mouse, Keyboard
- RAM : 8 GB

3.3.1 Database Requirements

- MySQL Database

3.4 Functional Requirements

- System must validate the previous block before commit block.
- User can access the data over the internet 24*7.
- If any block has changed by third party attacker or unauthorized user, it must show during transaction current blockchain is invalid.
- It can recover the invalid blockchain using other data nodes, with the help of majority of trustiness.

3.5 External Interface Requirements

3.5.1 User Interface

- User interface of this program is the common windows interface, nothing additional is required.
- The system user interface should be intuitive, such that 99.9% of all new system users are able to use system application without any assistance.

3.5.2 Hardware Interface

The hardware should have following specifications:

- Ability to read gallery
- Ability to exchange data over network
- Touch screen for convenience
- Keypad (in case touchpad not available)
- Continuous power supply
- Ability to connect to network
- Ability to take input from user
- Ability to validate user

3.5.3 Software Interface

- The software interfaces are specific to the target other user's proposed Application or software systems.

3.5.4 Communication Interface

- Our Project belongs to web based, so connecting user at online with request and response form. For that HTTP protocol we are going to use. That is provided by tomcat server 7/8.
- HTTP protocol: The Hypertext Transfer protocol is an application protocol for distributed, collaborative, and hypermedia information systems. HTTP is the foundation of data communication for the World Wide Web. Hypertext is structured text that uses logical links (hyperlinks) between nodes containing text.

3.6 Non-Functional Requirements

- System should be robustness
- The time required for processing the application will be less.
- The application must be scalable and reliable.
- Time saving i.e. time to generate detectors must be as minimum as possible.
- Accurate i.e. the accuracy of the anomaly detection system should be good as compared to proposed systems.

3.6.1 Performance Requirement

- System can produce results faster on 4GB of RAM.
- It may take more time for peak loads at main node.
- The system will be available 100% of the time. Once there is a fatal error, the system will provide understandable feed back to the user.

3.6.2 Safety Requirements

- Only administrators have access to the database of each individual user.
- All data will be backed-up every day automatically and also the system administrator can back-up the data as a function for him.
- This makes it easier to install and updates new functionality if required.
- For the safety purpose backup of the database must be required.

3.6.3 Security Requirements

- Our System is being developed in Python. Python is an Scripting programming language. Created by Guido van Rossum and first released in 1991, Java's design philosophy emphasizes code readability with its notable use of significant white-space.

- At the time of deploying this software user have to register to system.
- To use software user have to login and logout each time.

3.6.4 Software Quality Attributes

- **Runtime System Qualities:** Runtime System Qualities can be measured as the system executes.
- **Functionality:** The ability of the system to do the work for which it was intended.
- **Performance:** The response time, utilization, and throughput behavior of the system. Not to be confused with human performance or system delivery time.
- **Security:** A measure of systems ability to resist unauthorized attempts at usage or behavior modification, while still providing service to legitimate users.
- **Usability:** The ease of use and of training the end users of the system.
- **Sub qualities:** learn ability, efficiency, affect, helpfulness, control.
- **Interoperability:** The ability of two or more systems to cooperate at runtime.

3.6.5 Timeline Chart

Table 3.1: Timeline chart

Sr.No	Task Name	Duration	Start	Finish
1	Preliminary Phase	9 Day	27-09-2024	6-10-2024
2	Domain Selections	6 Day	07-10-2024	12-10-2024
3	Information Gathering	5 Day	12-10-2024	17-10-2024
4	Finalizing The Topic	7 Day	18-10-2024	25-10-2024
5	Requirement Gathering	7 Day	26-10-2024	02-11-2024
6	Analyzing the results	6 Day	02-11-2024	09-11-2024
7	Problem Identification	10 Day	10-11-2024	21-11-2024
8	Literature Review	12 Day	22-11-2024	02-12-2024
9	Requirement Specification	5 Day	03-12-2024	08-12-2024
10	Finalizing the requirement	4 Day	10-12-2024	15-12-2024
11	System Analysis	4 Day	16-12-2024	20-12-2024
12	Abstract Preparation	3 Day	21-12-2024	24-12-2024
13	Synopsis preparation	3 Day	21-12-2024	24-12-2024
14	UML diagram	3 Day	22-12-2024	24-12-2024
15	Architecture Design	5 Day	25-12-2024	31-12-2024
16	Coding	35 Day	01-01-2025	05-02-2025
17	Linking of Modules	2 Day	05-02-2025	07-02-2025
18	Testing	3 Day	07-02-2025	10-02-2025

Chapter 4

Design Of Software

This chapter introduces architecture of the system and module of the system. It also contain the registration, verification, authentication process and functioning of the system. DFD and UML diagram are explained in this chapter.

4.1 System Architecture

Figure 4.1 The proposed system integrates blockchain technology to create a secure and transparent pharmaceutical supply chain. At its core, a decentralized blockchain framework will record all transactions related to drug production, distribution, and sales, ensuring data integrity and traceability.

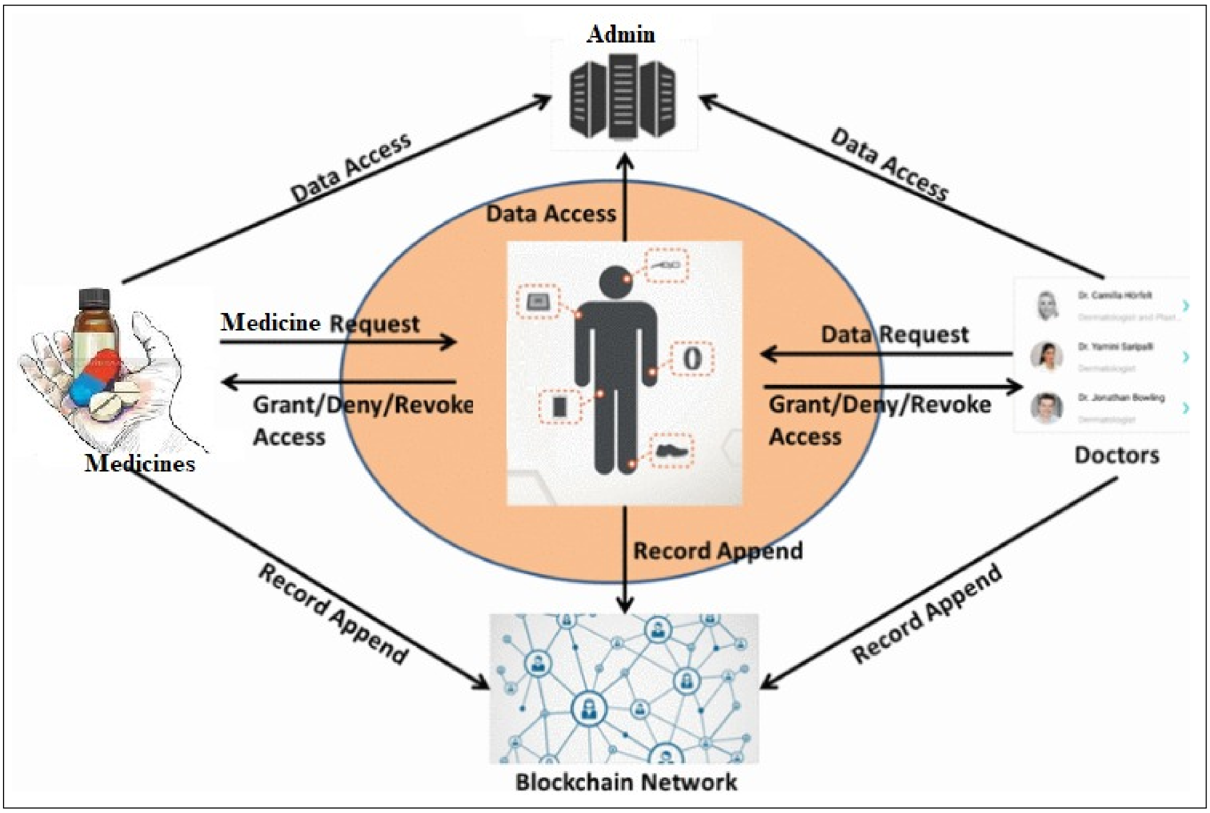

Figure 4.1: Proposed System Architecture

Smart contracts will automate agreements between stakeholders, such as manufacturers, distributors, and pharmacies, enforcing compliance and minimizing the risk of fraud. A secure user authentication system will ensure that only authorized participants, including manufacturers, distributors, hospitals, and patients, can access the platform.

The system will enable real-time tracking of pharmaceuticals, allowing stakeholders to verify the authenticity and status of products at any point in the supply chain. Additionally, a user-friendly interface will be provided for patients to confirm medication authenticity and for healthcare providers to efficiently manage prescriptions and inventory. Finally, the system will include reporting and analytics tools to generate insights on supply chain performance, counterfeit incidents, and compliance metrics, aiding in informed decision-making and enhancing safety. Overall, this comprehensive approach aims to significantly reduce the risk of counterfeit drugs entering the market and protect patient health.

The proposed system integrates blockchain technology to create a secure and transparent pharmaceutical supply chain. Key components of the system include:

System Architecture has this major blocks

- **Blockchain Framework:** A decentralized ledger to record all transactions related to drug production, distribution, and sales, ensuring data integrity and traceability.

- **Smart Contracts:** Automated agreements that enforce rules for transactions between stakeholders, such as manufacturers, distributors, and pharmacies, ensuring compliance and reducing fraud.

- **User Authentication:** A secure registration system for all participants, including manufacturers, distributors, hospitals, and patients, to ensure only authorized users access the system.

- **Real-Time Tracking:** A tracking system that monitors the movement of pharmaceuticals from production to delivery, allowing stakeholders to verify the authenticity and status of products at any point.

- **Patient and Healthcare Provider Interface:** A user-friendly dashboard that enables patients to verify medication authenticity and healthcare providers to manage prescriptions and inventory effectively.

- **Reporting and Analytics:** Tools for generating insights on supply chain performance, incidents of counterfeit drugs, and compliance metrics, aiding in decision-making and improving safety.

This proposed system aims to enhance the security and efficiency of the pharmaceutical supply chain, significantly reducing the risk of counterfeit drugs entering the market and protecting patient health.

The proposed system architecture aims to enhance the pharmaceutical supply chain's security and transparency using blockchain technology and smart contracts. The system comprises four main entities: Admin, Doctors, Patients (Users), and Blockchain Network, all interacting seamlessly to prevent counterfeit drugs and ensure legitimate medicine distribution.

1. **User-Centered Medicine Request and Access Control** At the core of the system is the User/Patient, who initiates the medicine request process. When a user needs a specific medicine, a Medicine Request is sent. This request is evaluated based on prescriptions and access rights, which are governed through smart contracts. The system checks the legitimacy of the user and the prescription by verifying data stored on the blockchain. Based on the verification, access is either granted, denied, or revoked dynamically, ensuring only authorized users can access genuine medicines.

2. **Doctor Involvement and Prescription Verification** Doctors play a critical role by generating authenticated prescriptions. They can request patient data, such as medical history or previous prescriptions, via smart contracts. The system ensures secure access to this data and records all actions immutably on the blockchain. When a doctor issues a prescription, it is immediately appended to the blockchain network, ensuring its authenticity and traceability. This prevents fake prescriptions from being used to acquire drugs illegally.

3. **Admin Oversight and Governance** The Admin entity acts as a system supervisor with elevated privileges. Admins manage stakeholder registration, system monitoring, and control access permissions. They have full data access and can intervene in access control mechanisms when required (e.g., in case of anomalies). Admin activities like data handling, permissions granting, and network configuration are also logged onto the blockchain, enhancing trust and auditability.

4. **Blockchain and Smart Contracts Execution** The Blockchain Network serves as the backbone of the system. All actions—including user requests, prescription approvals, medicine transactions, and permission grants—are managed via smart contracts. These contracts automate the flow without needing centralized intermediaries, reducing delay and fraud risks. Every transaction is appended to the ledger, enabling full traceability of medicine production, distribution, and delivery. This makes it nearly impossible for counterfeit drugs to enter the system unnoticed.

5. **Traceable and Tamper-Proof Supply Chain** Medicines flow from manufacturer to consumer with each stage—manufacturing, packaging, warehousing, shipping, and retail—recorded on the blockchain. Patients, doctors, and regulators can scan QR codes or tokens attached to medicine packs to trace their origin and verify authenticity in real time. This ensures tamper-proof tracking and deters illegal distribution.

This architecture offers a secure, transparent, and automated solution to combat counterfeit drugs while ensuring legitimate and efficient drug distribution through smart contracts and blockchain. It empowers all stakeholders with visibility, traceability, and accountability at every step of the pharma supply chain.

4.2 Data Flow Diagrams

A data flow diagram(DFD) is a graphical representation of the flow of data through information system, modeling it's process aspects. A DFD is often used as a preliminary step to create an overview of the system without going into great detail, which can later be elaborated.

1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.

2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.

3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.

4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

4.2.1 DFD 0 Level daigram

A context diagram is a top level(also known as "Level 0") data flow diagram. Figure 4.2 shows a Level 0 diagram.It only contain one process node("Process 0") that generalizes the function of the entire system in relationship to external entities.

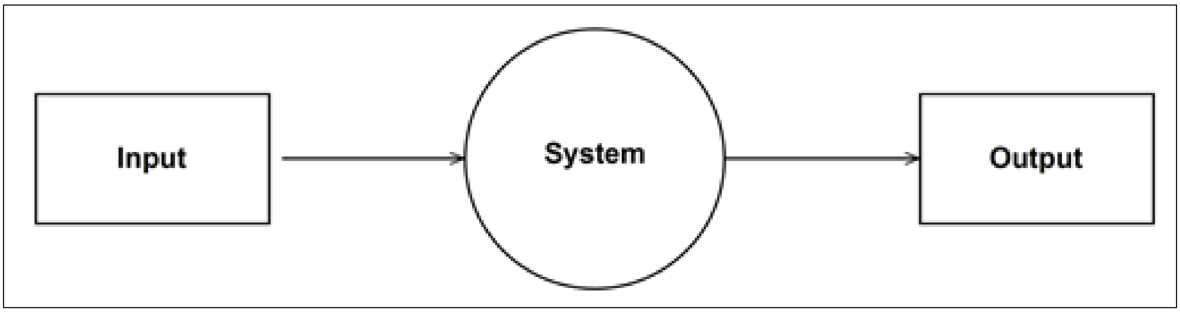

Figure 4.2: DFD Level 0 diagram

The Context Level DFD (Level 0) presents a high-level overview of the entire system, encapsulating its interaction with key external entities such as patients, doctors, administrators, and the blockchain network. The system is treated as a single process that handles user requests and responds accordingly. Patients request access to medicine, which the system verifies through a secure blockchain-based traceability check. Doctors interact with the system by uploading prescriptions and requesting access to patient or drug data. Admins manage the data access policies and oversee the flow of information. Meanwhile, the blockchain network records all approved interactions to ensure transparency and immutability. This level shows the overall flow of data without delving into system internals.

4.2.2 DFD 1 Level diagram

The Level 1 DFD shows how the system is divided into sub-system(processes),each of which deals with one or more of the data flows to or from an external agent and which together provide all of the functionality of the system as a whole.Figure 4.3 shows a Level 1 diagram.

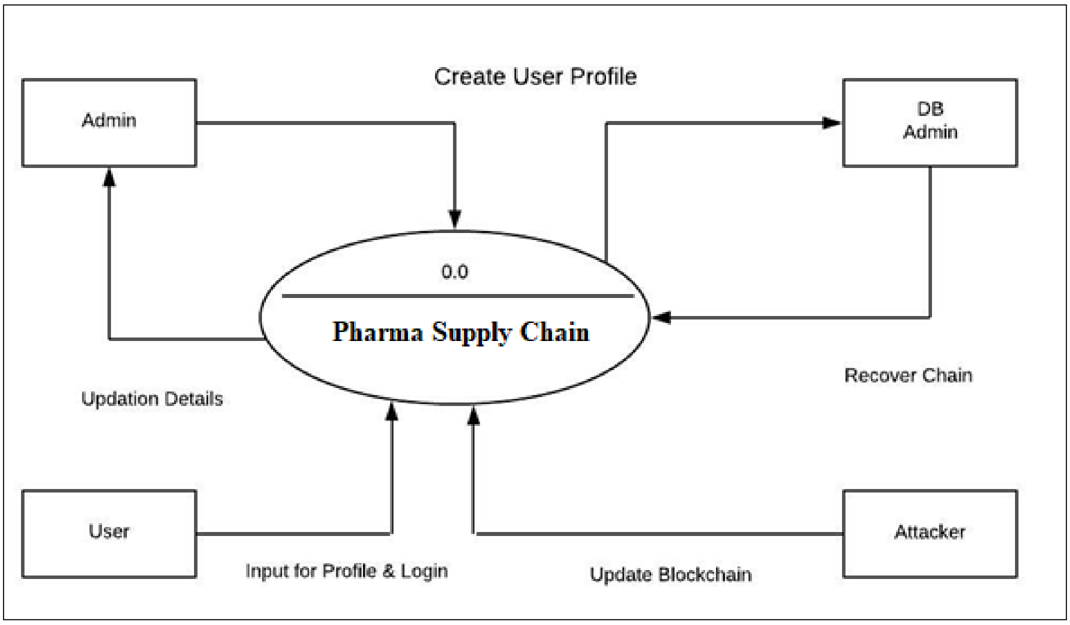

Figure 4.3: DFD Level 1 diagram

In the Level 1 DFD, the main system is broken down into multiple core modules to show how the data is processed internally. The first step is user authentication, where patients or doctors are verified using login credentials or digital signatures. Once authenticated, doctors upload patient prescriptions, which are validated through a prescription validation module. If valid, a smart contract is triggered to manage permissions. When a patient requests a specific drug, the system performs a drug trace verification to check its authenticity using blockchain records. If the medicine passes all validation checks, the smart contract execution module automatically grants access. All transactions and decisions are finally recorded by the blockchain record append process. Data is retrieved or stored in secure databases such as the user database, prescription ledger, and drug supply ledger.

A deeper view of the Level 2 DFD, particularly focusing on the smart contract execution module, shows a more refined breakdown. The system first checks for trigger conditions, such as a valid prescription or a drug request from a verified user. Next, it verifies the prescription hash, the doctor's identity, and the drug's origin. If all conditions are met, the smart contract autonomously grants access to the patient or updates the system about a genuine drug supply. In case of mismatch or tampering, access is rejected. Every step in this module is automatically logged and appended to the blockchain ledger for traceability, making it tamper-proof and secure.

Overall, these DFDs together ensure that each interaction in the pharmaceutical supply chain—from doctor's prescription to medicine dispensing—is validated through a transparent, decentralized, and automated process using blockchain and smart contracts. The system eliminates the possibility of counterfeit drug distribution by ensuring every drug's origin and journey is tracked and verified at each stage.

4.3 UML Diagrams

4.3.1 Use Case Diagram

Use case: UML stands for Unified Modeling Language. UML is a language for specifying visualization and documenting system. this is the step while developing any product after analysis. The goal from this is to produce a model of the entities involved in the project which later need to be build. The representation of the entities that are to be in the product being developed need to be designed.

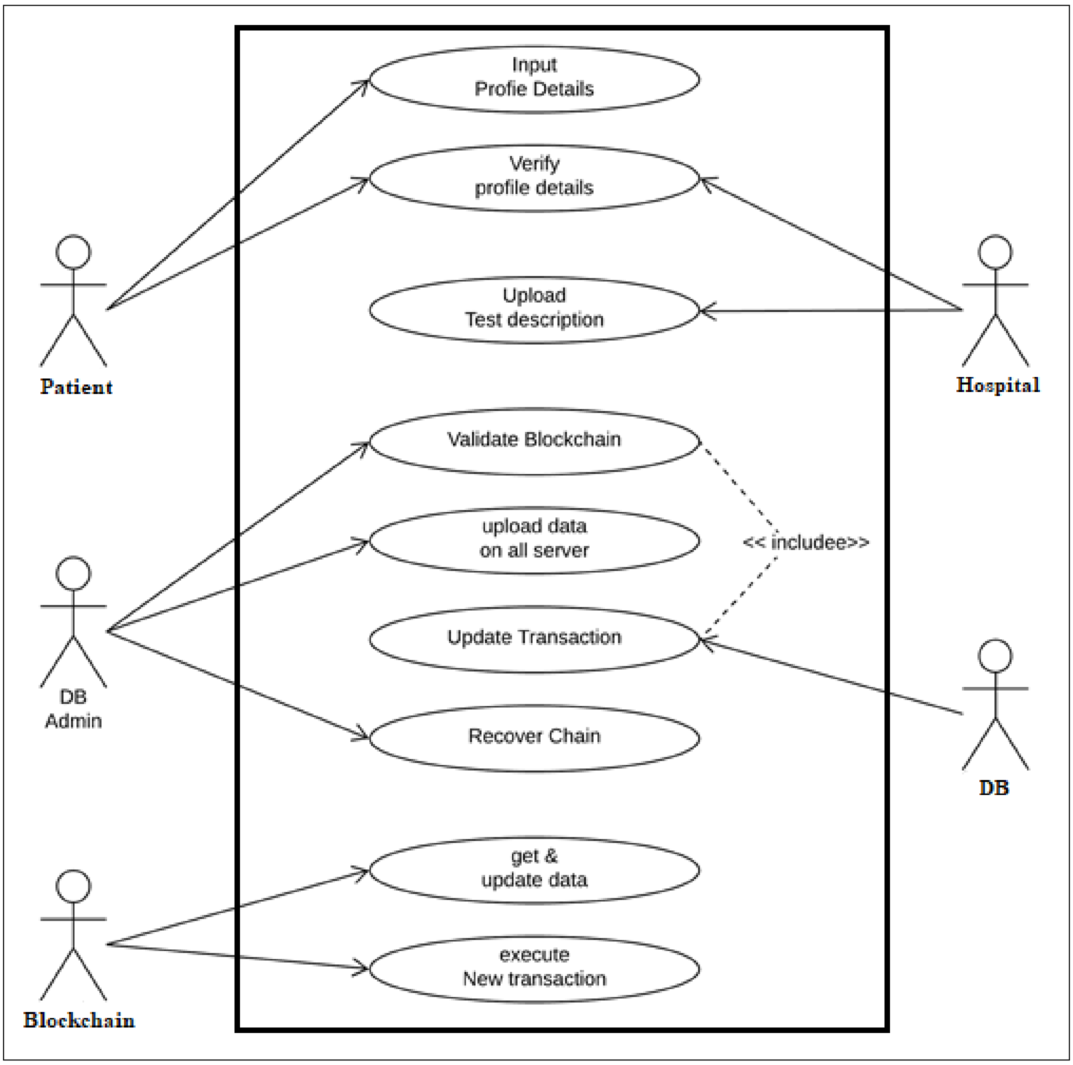

Figure 4.4: Use case Diagram

The use case diagram for the system titled **"Counterfeit Drug Prevention and Smart Contracts System using Pharma Supply Chain in Blockchain"** illustrates how different actors interact with the system components to ensure secure and traceable drug supply using blockchain. Key stakeholders include Patients, Hospitals, DB Admin, Database (DB), and Blockchain. Each actor is associated with specific actions that collectively prevent counterfeit drug distribution and ensure data integrity.

Actors and Use Cases

1. **Patient**

- **Input Profile Details:** The patient initiates the process by submitting personal and medical profile information.
- **Upload Test Description:** Patients can upload diagnostic reports or prescriptions that need to be validated and processed.

2. **Hospital**

- **Verify Profile Details:** The hospital validates the patient's submitted data for authenticity.
- **Upload Test Description:** In some cases, the hospital can also upload prescriptions and lab reports on behalf of the patient.

3. **DB Admin**

- **Validate Blockchain:** The DB Admin ensures that the current block structure is intact and unaltered.
- **Upload Data on All Servers:** Ensures decentralized storage and synchronization across all nodes.
- **Update Transaction:** Updates blockchain with new verified medical or drug-related transactions.
- **Recover Chain:** Recovers any lost or corrupted data by retrieving from other verified nodes.

4. **DB (Database)**

- **Update Transaction:** Works in collaboration with DB Admin to perform CRUD operations on verified records.

5. **Blockchain**

- **Get & Update Data:** Retrieves and updates chain records as validated transactions are performed.
- **Execute New Transaction:** Adds new blocks after transactions are approved, ensuring immutability.

System Flow Summary

The process starts with data entry by patients or hospitals. Once submitted, the DB Admin validates and verifies the integrity of this data using blockchain mechanisms. Once verified, transactions are logged, and the chain is updated. In case of discrepancies or data loss, the DB Admin uses the recovery mechanism to restore accurate data from other distributed nodes. Blockchain ensures transparent execution of all new transactions.

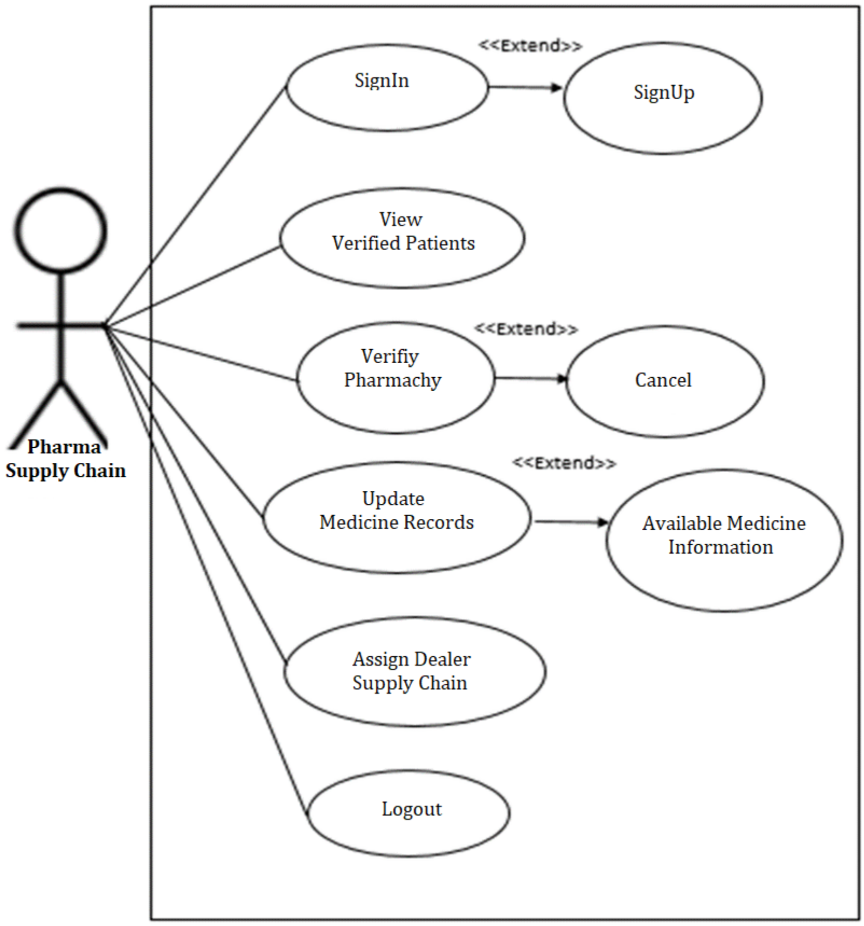

Figure 4.5: Use Case Diagram - Pharma Supply Chain

Use Case Diagram - Pharma Supply Chain

This use case diagram represents the specific interactions that the **Pharma Supply Chain** actor performs in the system titled *"Counterfeit Drug Prevention and Smart Contracts System using Pharma Supply Chain in Blockchain"*. The goal is to ensure end-to-end transparency and traceability of drug data, verification, and distribution through blockchain smart contracts.

Actor: Pharma Supply Chain

The Pharma Supply Chain actor is responsible for maintaining and updating drug-related records, managing pharmacies, and ensuring verified dealer assignments. This actor also interacts with system features such as medicine record updates and supply chain dealer management.

Use Cases

- **SignIn:** Allows pharma officials to log into the system.
- **SignUp (Extended Use Case):** A prerequisite for new users to register before

signing in.

- **View Verified Patients:** Enables viewing of patients whose medical data has been validated in the blockchain.
- **Verify Pharmacy:** Pharma supply authorities validate the authenticity of registered pharmacies.
  - **Cancel (Extend):** The verification process can be canceled if validation fails or inconsistencies are found.
- **Update Medicine Records:** Allows authorized personnel to input or edit existing drug information in the blockchain.
  - **Available Medicine Information (Extend):** Access current availability and status of medicines during record updates.
- **Assign Dealer Supply Chain:** Assign dealers for distributing verified medicines across the supply chain.
- **Logout:** Securely exits the session to ensure no unauthorized actions.

4.3.2 Sequence Diagram

Sequence diagram and collaboration diagram are call INTERACTION DIAGRAM. An interaction diagram shows an interaction, consisting of set of object and their relationship including the message that may be dispatch among them sequence diagram simply depicts interaction between objects in a sequential order i.e. the order in which these interactions take place. We can also use the terms event diagrams or event scenarios to refer to a sequence diagram. Sequence diagrams describe how and in what order the objects in a system function. Figure 4.6 shows a Sequence Diagram.

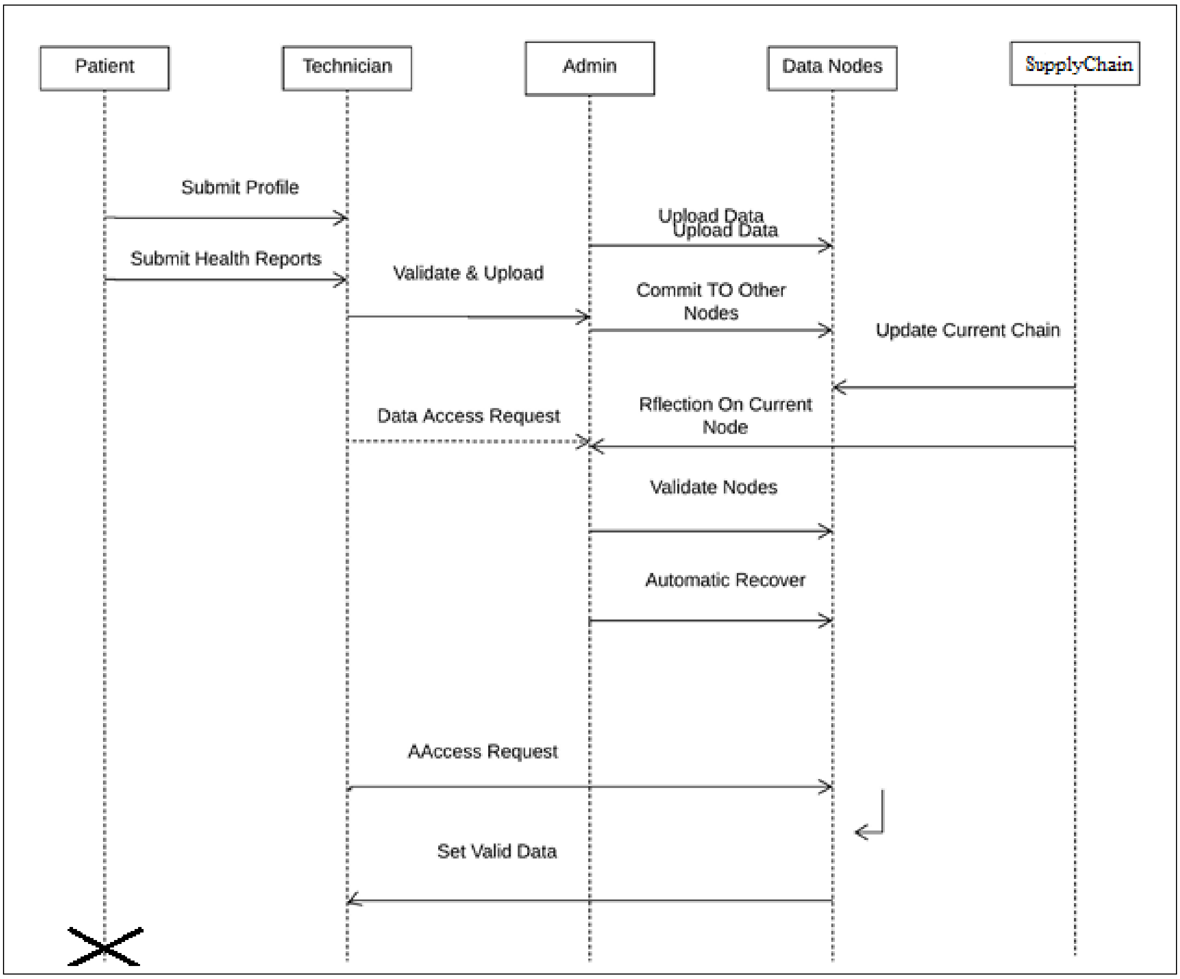

Figure 4.6: Sequence Diagram

4.3.3 Class Diagram

Class diagram are the most common diagrams used in UML. Class diagram consists of classes, interfaces, association and collaboration. It basically represent the object oriented view of a system which is static in nature.Figure 4.7 shows a class diagram.

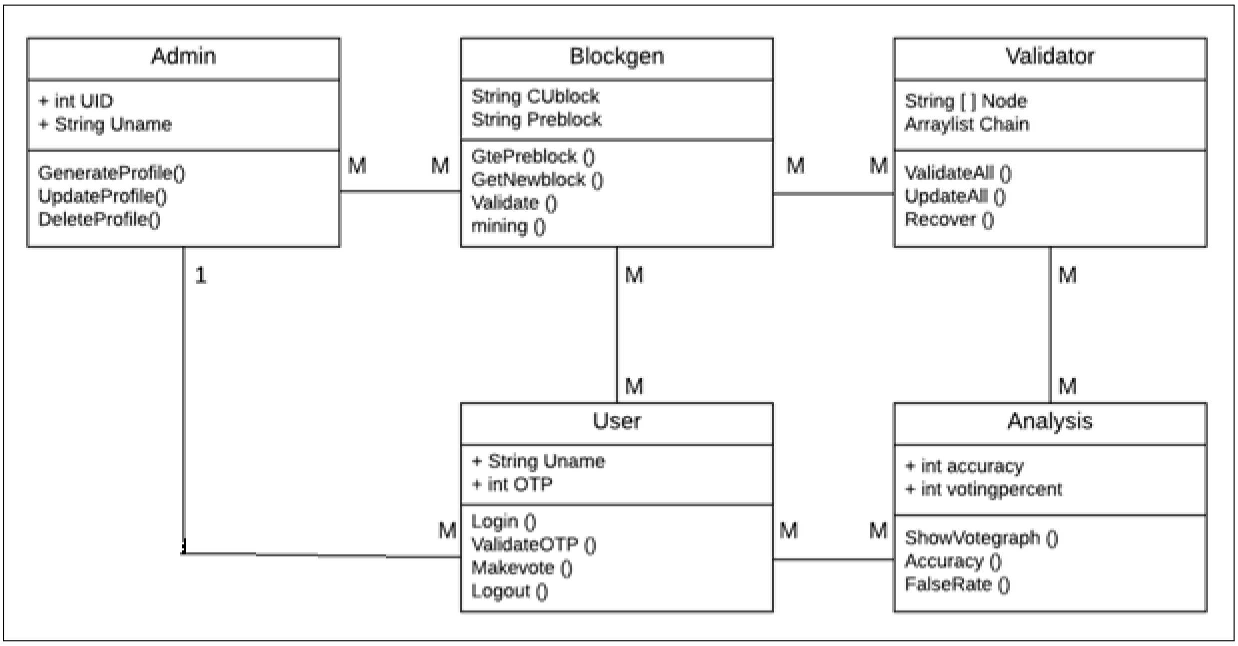

Figure 4.7: Class Diagram

The class diagram represents the modular structure of the system designed for **Counterfeit Drug Prevention and Smart Contracts System using Pharma Supply Chain in Blockchain**. Each class encapsulates entities such as Admins, Users, Blockchain components, Validators, and Analytical modules. The diagram defines the attributes, operations, and relationships among these entities, reflecting how real-world interactions are managed securely on the blockchain.

Class Descriptions

1. Admin

Attributes:

- `UID` – Unique identifier for the Admin.
- `Uname` – Username of the Admin.

Methods:

- `GenerateProfile()` – Creates a profile for a new user or doctor.
- `UpdateProfile()` – Edits existing profile details.
- `DeleteProfile()` – Deletes a user or doctor profile.

The Admin plays a crucial role in managing user accounts and maintaining system oversight.

2. User

Attributes:

- `Uname` – User's login name.
- `OTP` – One-time password for authentication.

Methods:

- `Login(), ValidateOTP(), Makevote(), Logout()`

Users represent real-world stakeholders such as doctors, pharmacists, and patients who access the system to authenticate medicine requests.

3. Blockgen (Block Generator)

Attributes:

- `CuBlock` – Current block being processed.
- `PreBlock` – Reference to the previous block.

Methods:

- `GetPreblock(), GetNewblock(), Validate(), Mining()`

Blockgen ensures that all transactions, including medicine tracking and prescription approvals, are securely recorded on the blockchain.

4. Validator

Attributes:

- `Node[]` – Array of validator nodes.
- `Chain` – The blockchain ledger being validated.

Methods:

- `ValidateAll(), UpdateAll(), Recover()`

Validators authenticate blocks before they are permanently added to the blockchain, ensuring trust and data integrity.

5. Analysis

Attributes:

- `accuracy` – Model or system accuracy.
- `votingpercent` – Percentage score from voting modules.

Methods:

- `ShowVoteGraph(), Accuracy(), FalseRate()`

This class helps with performance auditing and counterfeit detection reporting, providing insights to both system admins and authorities.

Class Relationships

- The `Admin` class has a one-to-many relationship with `User`, enabling a single Admin to manage multiple Users.
- All core components (`Admin, User, Validator, Analysis`) interact with `Blockgen`, highlighting its role as the blockchain data handler.
- Validators and Analysis classes ensure the integrity of data and provide auditing and feedback loops for quality control.

Real-Time Use Case

When a doctor logs in and submits a prescription, the transaction is first authenticated and then passed to the Blockgen. Validator nodes verify its legitimacy, and once confirmed, the block is appended to the chain. The Admin may oversee and adjust access, while the Analysis class monitors the accuracy and identifies potential counterfeit patterns.

4.3.4 Activity Diagram

Activity diagram are the graphical representation of work flows of step wise activities and action with support for choice, iteration and concurrency. In this Unified Modeling Language, Activity diagram can be used to describe the business and operational step-by-step work flows of component in a system. An activity diagram are constructed from limited number of shapes, connected with arrows. The most important shape type: Arrows run from the start towards the end represent the order in which activity has done.

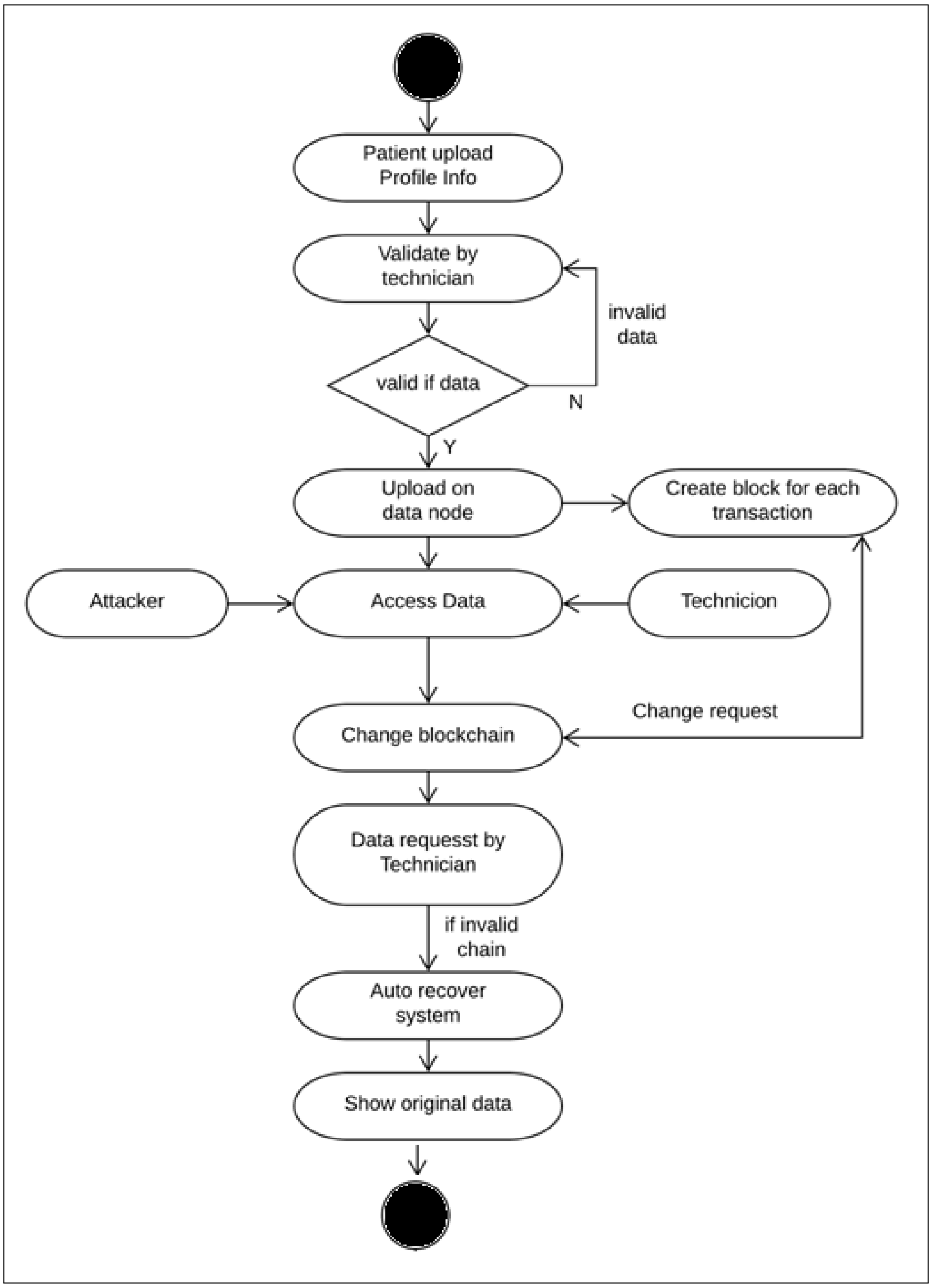

Figure 4.8: Activity Diagram

The activity diagram illustrates the dynamic workflow of data validation, transaction logging, and security checks within the system. This flow plays a crucial role in preventing counterfeit drugs by ensuring that only validated data is uploaded and transactions are securely stored in the blockchain.

Real-time Process Explanation

- The process begins when a **Patient uploads profile information**.
- A **Technician validates** the uploaded data.
- If the data is **invalid**, the process is terminated and returned for correction.
- If the data is **valid**, it is uploaded to the **data node**, and a **block is created** for each transaction.
- Meanwhile, if an **attacker attempts to access or change the data**, the system stores a change record.
- Any **change request by the technician** triggers a verification.
- During data access by technicians, the system checks whether the **blockchain is valid**.
- If the chain is found to be **tampered**, the system activates an **auto-recover mechanism** that restores the original data from the verified copy.
- Finally, the system displays the **original untampered data** to the technician.

4.3.5 Deployment Diagram

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed. So deployment diagrams are used to describe the static deployment view of a system. Deployment diagrams consist of nodes and their relationships. Deployment diagrams are used for describing the hardware components where software components are deployed. Component diagrams and deployment diagrams are closely related. Component diagrams are used to describe the components and deployment diagrams shows how they are deployed in hardware.

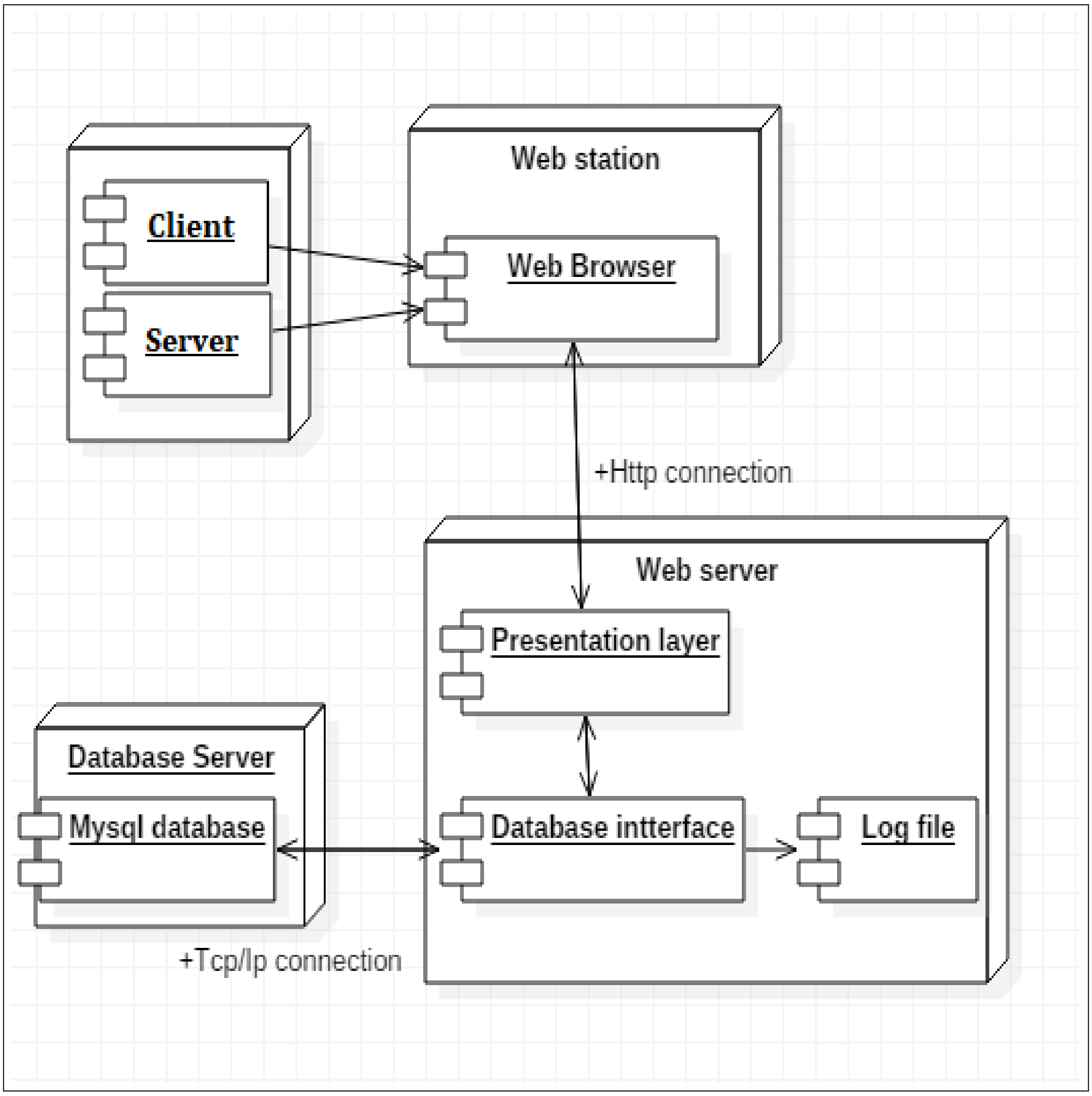

Figure 4.9: Deployment Diagram

4.4 Modules of System

4.4.1 Patient Module

Allows patients to verify the authenticity of their medications using a unique identifier or QR code. Provides access to information about drug origins, ensuring transparency in the pharmaceutical supply chain. Enables reporting of suspected counterfeit drugs, enhancing community vigilance.

- **Input:** Patient identification details, prescription information, and drug requests.
- **Processing:** Validates the patient's identity, checks prescription validity, and retrieves drug availability from the supply chain.
- **Output:** Confirmation of drug availability, purchase options, and notifications about counterfeit risks.

4.4.2 Hospital Module

Facilitates hospitals to track and manage their medication inventory using blockchain technology for real-time updates. Allows healthcare providers to access verified information about drug suppliers and batch details.
Supports integration with electronic health records (EHR) to ensure that prescribed medications are legitimate and safe.

- **Input:** Patient records, drug inventory data, and procurement requests.
- **Processing:** Manages patient treatment data, monitors inventory levels, and places orders for drugs from suppliers.
- **Output:** Inventory reports, procurement status updates, and alerts for low stock or counterfeit alerts.

4.4.3 Admin Module

Centralizes oversight of the entire supply chain, allowing administrators to monitor transactions and verify drug authenticity. Manages user roles and permissions for patients, hospitals, and pharmaceutical suppliers to ensure secure access. Generates reports and analytics on drug transactions, helping to identify trends and potential fraud.

- **Input:** User management details, system configurations, and compliance reports.
- **Processing:** Administers user accounts, oversees system security, and generates compliance reports to ensure regulatory adherence.
- **Output:** User access logs, system performance reports, and compliance alerts.

4.4.4 Pharmaceutical Supply Chain Module

Tracks the entire lifecycle of medications from manufacturing to distribution, ensuring full traceability. Implements smart contracts that automatically validate transactions and trigger alerts for any discrepancies. Integrates with manufacturers, distributors, and pharmacies to maintain a secure and transparent supply chain.

- **Input:** Drug production details, distribution logistics, and transaction records.
- **Processing:** Tracks drug movement through the supply chain, verifies authenticity using blockchain, and manages supplier relationships.
- **Output:** Supply chain transparency reports, drug authenticity confirmations, and alerts for any discrepancies or counterfeit risks.

4.4.5 Regulatory Compliance Module

Ensures that all stakeholders comply with relevant regulations and standards regarding drug safety and authenticity. Provides tools for regulatory authorities to access data and perform audits more efficiently. Facilitates communication between stakeholders and regulatory bodies to streamline compliance processes.

4.4.6 Pharma Supplier Module

Allows suppliers to register and verify their credentials on the blockchain, enhancing trust within the supply chain. Provides tools for updating inventory and tracking shipment statuses in real-time.

Enables transparent communication with hospitals and pharmacies regarding drug availability and authenticity. These modules work together to create a comprehensive system that enhances the safety, transparency, and efficiency of the pharmaceutical supply chain, significantly reducing the risks associated with counterfeit drugs.

Chapter 5

Implentation Details

5.1 Requirement Gathering

Requirement analysis is for transformation of operational need into software description, software performance parameter, and software configuration through use of standard, iterative process of analysis and trade-off studies for understanding what the customer wants analyzing need, assessing feasibility, negotiating a reasonable solution validating the specification and managing the requirements.
The purpose of System Requirements Analysis is to obtain a thorough and detailed understanding of the business need as defined in Project Origination and captured in the Business Case, and to break it down into discrete requirements, which are then clearly defined, reviewed and agreed upon with the Customer Decision Makers. During System Requirements Analysis, the framework for the application is developed, providing the foundation for all future design and development efforts. System Requirements Analysis can be a challenging phase, because all of the major Customers and their interests are brought into the process of determining requirements. The quality of the final product is highly dependent on the effectiveness of the requirements identification process.

1. **Module 1: Blockchain:**
   The concept of blockchain was proposed by Satoshi Nakamoto in 2008. Blockchain is an online ledger that provides decentralized and transparent data sharing. With distributed recordings, all transaction data (stored in nodes) are compressed and added to different blocks. Data of various types are distributed in distinct blocks, enabling verifications to be made without the use of intermediaries. All the nodes then form a blockchain with timestamps. The data stored in each block can be verified simultaneously and become inalterable once entered. The whole process is open to the public, transparent, and secure [8].

2. **Module 2: Custom Blockchain:**
   The infrastructure is developed and maintained by both Ethereum and those developers. The major characteristics of Ethereum are as follows:
   1) Incorruptible: third-parties are not able to modify any data;
   2) Secure: errors derived from personnel factors are avoided because the decentralized applications are maintained by entities rather than individuals;

3) Permanent: blockchain does not cease to operate even if an individual computer or server crashes.

3. **Module 3: Smart Contracts:**
   Smart contracts were first proposed by Nick Szabo in the early 1990s. He explained that a smart contract enabled computers to execute transaction clauses. As blockchain has become popular, smart contracts have received increased attention. Smart contracts are the main feature of Custom, a blockchain platform founded in 2015. A smart contract is "a digital contract that is written in source code and executed by computers, which integrates the tamper-proof mechanism of blockchain" [6]. Smart contracts can be created using the Custom blockchain. Developers are able, according to their needs, to specify any instruction in smart contracts; develop various types of applications, including those that interact with other contracts; store data; and transfer Custom.
   Additionally, smart contracts that are deployed in blockchain are copied to each node to prevent contract tampering. With related operations executed by computers and services provided by Custom, human error can be reduced to avoid disputes regarding such contracts. Smart contracts are mostly used in voting system [7] and crypto-currency applications. The high-level programming languages used for writing smart contracts are mainly Solidity, Serpent, and LLL [12]. Currently, most developers employ Solidity to write smart contracts and compile the instructions into byte code for the EVM to execute. Certain costs are incurred when developers create smart contracts.

4. **Module 4: SHA Hash Generation:**
   The SHA-256 algorithm is a hashing algorithm that performs on data in one-way and Ron Rivist develops it. It is an evolution of previous algorithms such as SHA 0, SHA 1, SHA 256, SHA 384. Hashing is also known as compression or message summary function, which takes the entire variable length and changes it into a binary sequence of fixed length.

5.2 Tools and Technologies Used

- **JAVA:** Java is one of the most popular and widely used programming language and platform. A platform is an environment that helps to develop and run programs written in any programming language.
  Java is fast, reliable and secure. From desktop to web applications, scientific supercomputers to gaming consoles, cell phones to the Internet, Java is used in every nook and corner.
  Java is a programming language and computing platform first released by Sun Microsystems in 1995. There are lots of applications and websites that will not work unless you have Java installed, and more are created every day. Java is fast, secure, and reliable. From laptops to datacenters, game consoles to scientific supercomputers, cell phones to the Internet, Java is everywhere!
  Java is a general-purpose, concurrent, object-oriented, class-based, and the runtime environment(JRE) which consists of JVM which is the cornerstone of the Java platform. This blog on What is Java will clear all your doubts about

why to learn java, features and how it works.

- **XAMPP:** XAMPP stands for Cross-Platform (X), Apache (A), MySQL (M), PHP (P) and Perl (P). It is a simple, lightweight Apache distribution that makes it extremely easy for developers to create a local web server for testing purposes. Everything you need to set up a web server – server application (Apache), database (MySQL), and scripting language (PHP) – is included in a simple extractable file. XAMPP is also cross-platform, which means it works equally well on Linux, Mac and Windows. Since most actual web server deployments use the same components as XAMPP, it makes transitioning from a local test server to a live server is extremely easy as well. Web development using XAMPP is especially beginner friendly.

- **JDK:** The Java Development Kit (JDK) is an implementation of either one of the Java Platform, Standard Edition, Java Platform, Enterprise Edition, or Java Platform, Micro Edition platforms released by Oracle Corporation in the form of a binary product aimed at Java developers on Solaris, Linux, macOS or Windows. The JDK includes a private JVM and a few other resources to finish the development of a Java Application. Since the introduction of the Java platform, it has been by far the most widely used Software Development Kit (SDK).[citation needed] On 17 November 2006, Sun announced that they would release it under the GNU General Public License (GPL), thus making it free software. This happened in large part on 8 May 2007, when Sun contributed the source code to the OpenJDK.

- **Apache:** Apache is the actual web server application that processes and delivers web content to a computer. Apache is the most popular web server online, powering nearly 54 percent of all websites.

- **MySQL:** Every web application, howsoever simple or complicated, requires a database for storing collected data. MySQL, which is open source, is the world's most popular database management system. It powers everything from hobbyist websites to professional platforms like WordPress. You can learn how to master PHP with this free MySQL database for beginners course.

5.3 Algorithm Details

5.3.1 Algorithm /Pseudo Code

1. Shamir Secret Sharing:

In cryptography, secret sharing schemes are schemes that split shares of a given secret among a set of trusted participants. This secret could be a very important piece of information that may be needed in the future but needs to be kept private and secure. These shares are completely useless on their own, however, when put together reconstruct the secret and make it apparent. As a thought experiment;

think of secret sharing schemes as a puzzle where the puzzle pieces are split among ten players but are completely blank. The image that the puzzle creates is only visible once all the pieces are put together.
The kind of data that is best suited for a secret sharing algorithm is information that must be kept absolutely private, but must also be stored securely and never lost. Typically, you'd use secret sharing for access keys to accounts with highly sensitive information in them. The goal is to spread the key out from one geographic location into multiple—so that in order to compromise a system, you'd need to compromise devices in distinct locations first. There are multiple kinds of secret sharing algorithms. The one we'll be discussing is Shamir's Secret Sharing.

```
    <shamirsecret = ShamirSecret.new(2,
''In the name of Adi Shamir")>
    s1 = shamirsecret.compute_share(1)
    s2 = shamirsecret.compute_share(2)
    s3 = shamirsecret.compute_share(3)

    shamirsecret = nil

    shamirsecret.recover_secretdata([s1,s3])

shamirsecret = ShamirSecret.new(2, "In the name of Adi Shamir")
s1 = shamirsecret.compute_share(1)
s2 = shamirsecret.compute_share(2)
s3 = shamirsecret.compute_share(3)

# Make sure our share is valid
shamirsecret.is_valid_share(s1)
=> true

# Change a byte of share
s1[1][0] = s1[1][0] + 1 % 256

# The share is not valid anymore
shamirsecret.is_valid_share(s1)
=> false

# Secret is corrupted with wrong share
shamirsecret = nil
shamirsecret = ShamirSecret.new(2)
shamirsecret.recover_secretdata([s1,s3])
=> "\xC6n the name of Adi Shamir"
```

2. SHA1: (Hash)
SHA-1 or Secure Hash Algorithm 1 is a cryptographic hash function which takes an input and produces a 160-bit (20-byte) hash value. This hash value is known as

a message digest. This message digest is usually then rendered as a hexadecimal number which is 40 digits long. It is a U.S. Federal Information Processing Standard and was designed by the United States National Security Agency.
SHA-1 is now considered insecure since 2005. Major tech giants browsers like Microsoft, Google, Apple and Mozilla have stopped accepting SHA-1 SSL certificates by 2017.
To calculate cryptographic hashing value in Java, MessageDigest Class is used, under the package java.security.
These algorithms are initialized in static method called getInstance(). After selecting the algorithm the message digest value is calculated and the results are returned as a byte array. BigInteger class is used, to convert the resultant byte array into its signum representation. This representation is then converted into a hexadecimal format to get the expected MessageDigest.

```
MessageDigest md = MessageDigest.getInstance("SHA-1");

    byte[] messageDigest = md.digest(input.getBytes());

    BigInteger no = new BigInteger(1, messageDigest);

    String hashtext = no.toString(16);

    while (hashtext.length() < 32) {
        hashtext = "0" + hashtext;
}

    return hashtext;
}

catch (NoSuchAlgorithmException e) {
    throw new RuntimeException(e);
}
```

5.3.2 Algorithm

Algorithm 1:- Protocol for Peer Verification:
Input: User get IP address, User Transaction TID, **Output:** Enable IP address or current query if any connection is valid
Step 1 : User generate the any transaction DDL, DML or DCL query
Step 2 : Get current IP address
For each (read IP into IP address)
If (connection(IP) equals(true))
Flag true
Else
Flag false End for
Step 4 : if (Flag == true)
Peer to Peer Verification valid Else
Peer to Peer Verification Invalid End if

End for

Algorithm 2:- Hash Generation:
Input: Genesis block, Previous hash, data d, **Output:** Generated hash H according to given data
Step 1 : Input data as d
Step 2 : Apply SHA 256 from SHA family
Step 3 : CurrentHash= SHA256(d)
Step 4 : Retrun CurrentHash

Algorithm 3:- Mining Algorithm for valid hash creation:
Input: Hash Validation Policy P[], Current Hash Values hash Val
Output: Valid hash
Step 1 : System generate the hash Val for ith transaction using Algorithm 1
Step 2 : if (hash Val.valid with P[])
Valid hash Flag =1 Else Flag=0
Mine again randomly
Step 3 : Return valid hash when flag=1

5.4 Mathematical Model

1. System Description:
A System has represented by a 5-different phases, each phase works with own dependency System S = (Q, Σ, δ, q0, F) where :

- Q is a finite set of states.
- Σ is a finite set of symbols called the alphabet.
- Δ is the transition function where δ : Q x Σ → Q
- q0 is the initial state from where any input is processed (q0 ϵ Q).
- F is a set of final state/states of Q (F ⊆ Q).

2. Analysis:-
All t(n) policies will return 1 then from training patterns and it generate the similarity weight of fitness function of specific rules.
Q = {VaiSet[i=0.......n]} set of generated attribute of various images as initial set.
Σ = {data conversion, saveinDB}
Δ = {_CorrecltyclassifiedInstnaces*100 / SumF(x)}
q0 = {First event generated by sensor function Σ i=0 }
F = {Generated report according to class [a,b,c.....n]}

3. Intersection of two sets:-

4. Success Conditions:
Safe and Unsafe Transaction Detected using Concensus algorithm and only the safe

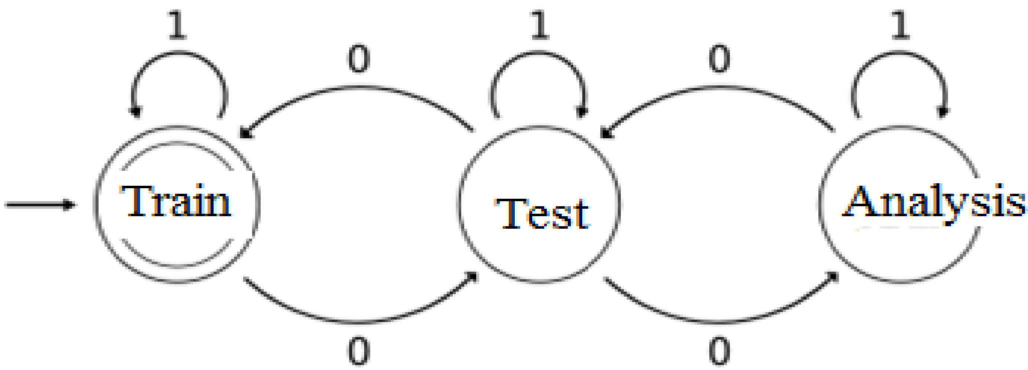

Figure 5.1: Mathematical Model

transaction sent to user.

5. Failure Condition:

If IoT does not working hardware or system properly this will be a case of failure.

	0	1
S1	S2	S1
S2	S1	S2

Table 5.1: State Transition Table

The state S1 represents that there has been an even number of 0s in the input so far, while S2 signifies an odd number. A 1 in the input does not change the state of the automaton. When the input ends, the state will show whether the input contained an even number of 0s or not. If the input did contain an even number of 0s, M will finish in state S1, an accepting state, so the input string will be accepted.

Time Complexity of system:

The runtime complexity of Algorithm 1 and Procedure 1 is nLog(n), where —D— is the number of records in the input dataset D and —I— the size of the item universe. The main computational cost comes from the distribution of records from a partition to its sub-partitions. The complexity of distributing the records for a single partition operation is O(—D—) because a partitioning can affect at most —D— records. According to algorithms, the maximum number of partitioning needed for the entire process is the number of internal nodes in the taxonomy tree H. For a taxonomy tree with a fan-out f 2, the number of internal nodes is —I—1 f1. Therefore, the overall complexity of our approach is nLog(n).

Chapter 6

Results

6.1 Project Module Implementation Screenshots

This chapter introduce us about the prototype model (GUI) in which it known us how project look like and implementation detail of first module

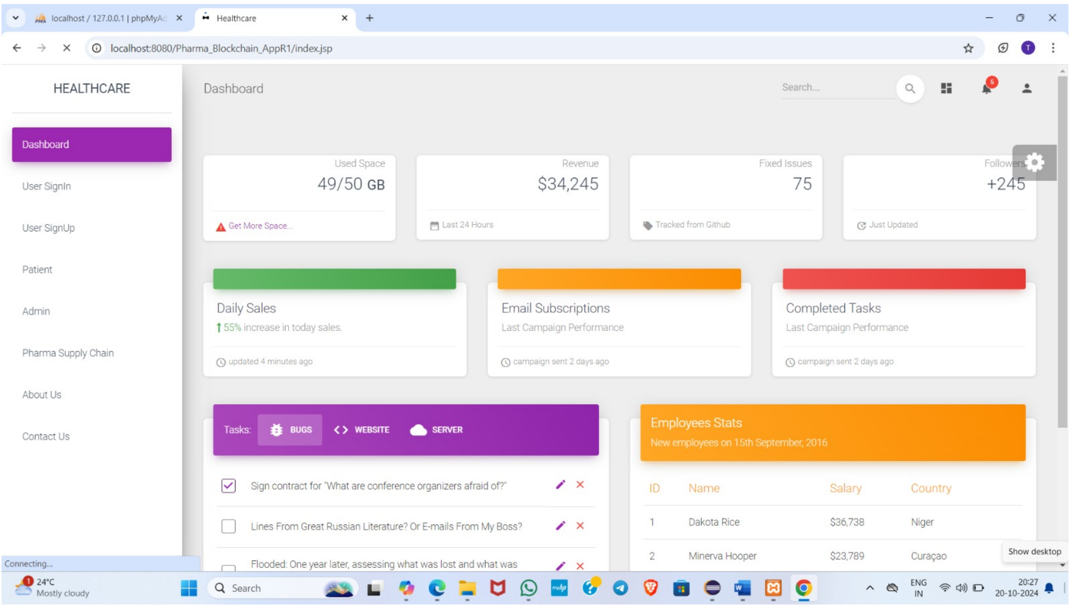

Figure 6.1: Home page

Make that the pharmaceutical supply chain is transparent, effective, and traceable. Using real-time data and cognitive analytics, our platform assists manufacturers, distributors, and retailers with inventory management, shipment tracking, and the fight against counterfeit medications.

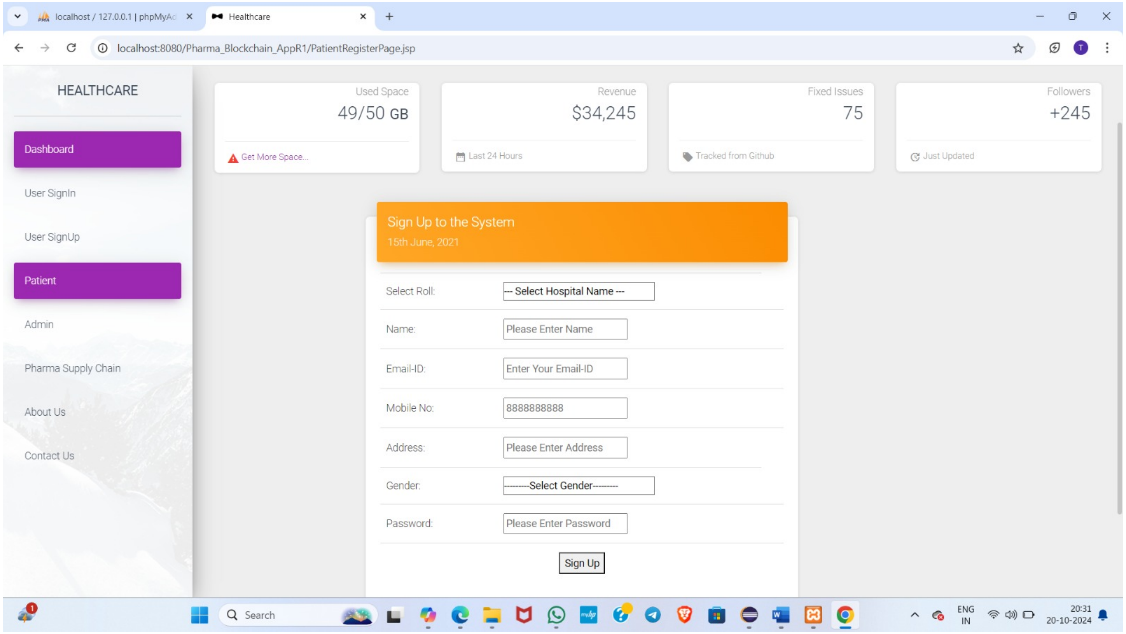

Figure 6.2: Registration Page

In the module shown in Figure 6.2, the user must enter all the necessary credentials required for the initial registration. Post successful registration the user data is stored into the application database and then he/she can easily log into the application.

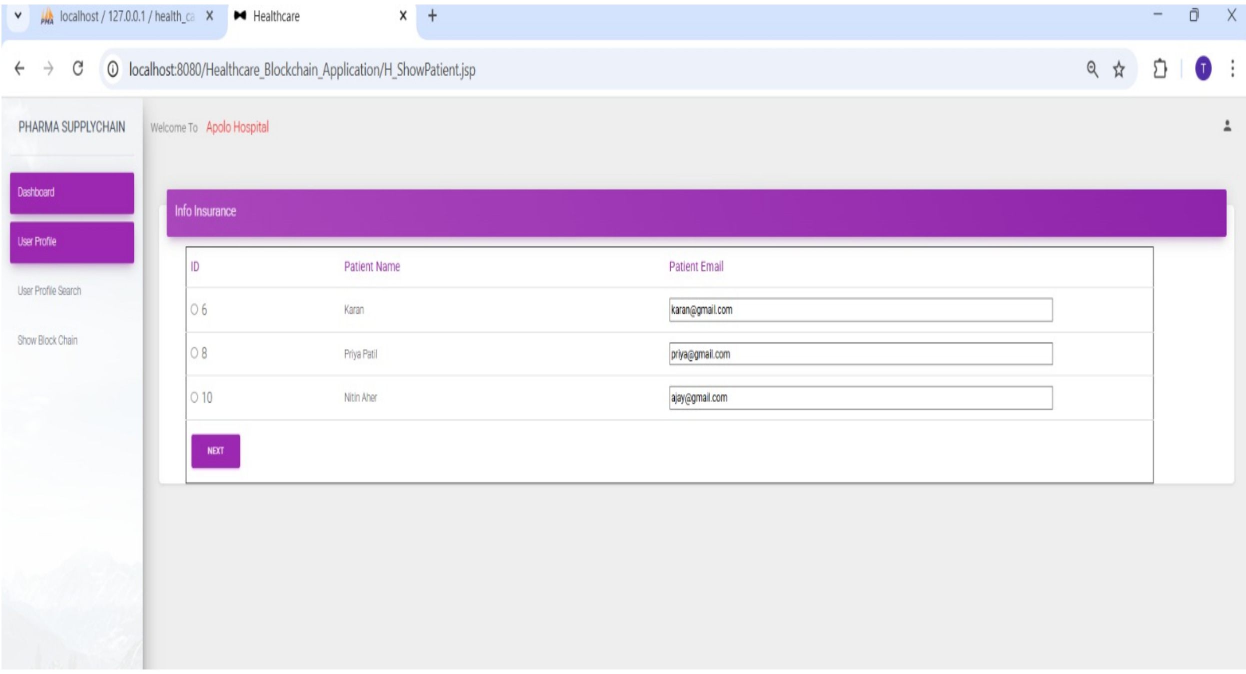

Figure 6.3: Running Project Snapshot

In the module shown in Figure 6.3, the user must enter all the necessary credentials required for the initial login authentication. Post successful login the user data is access into the application database and then he/she can easily log into the application.

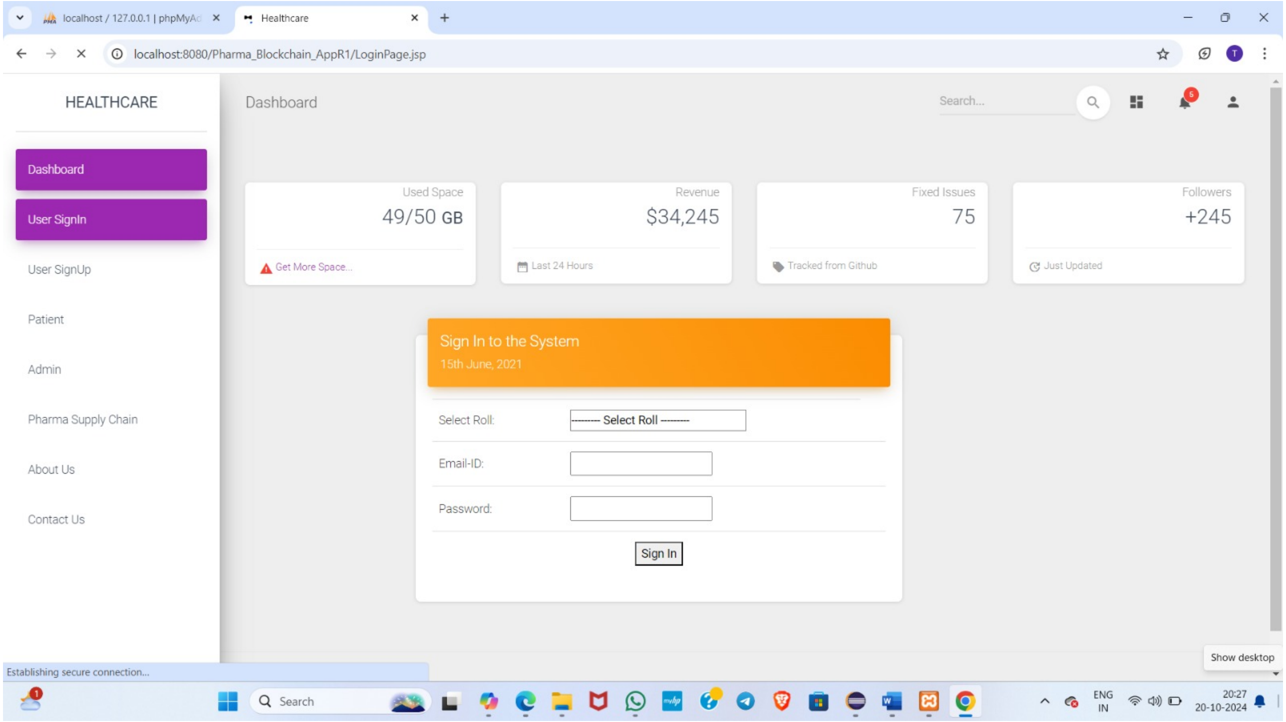

Figure 6.4: Login Page

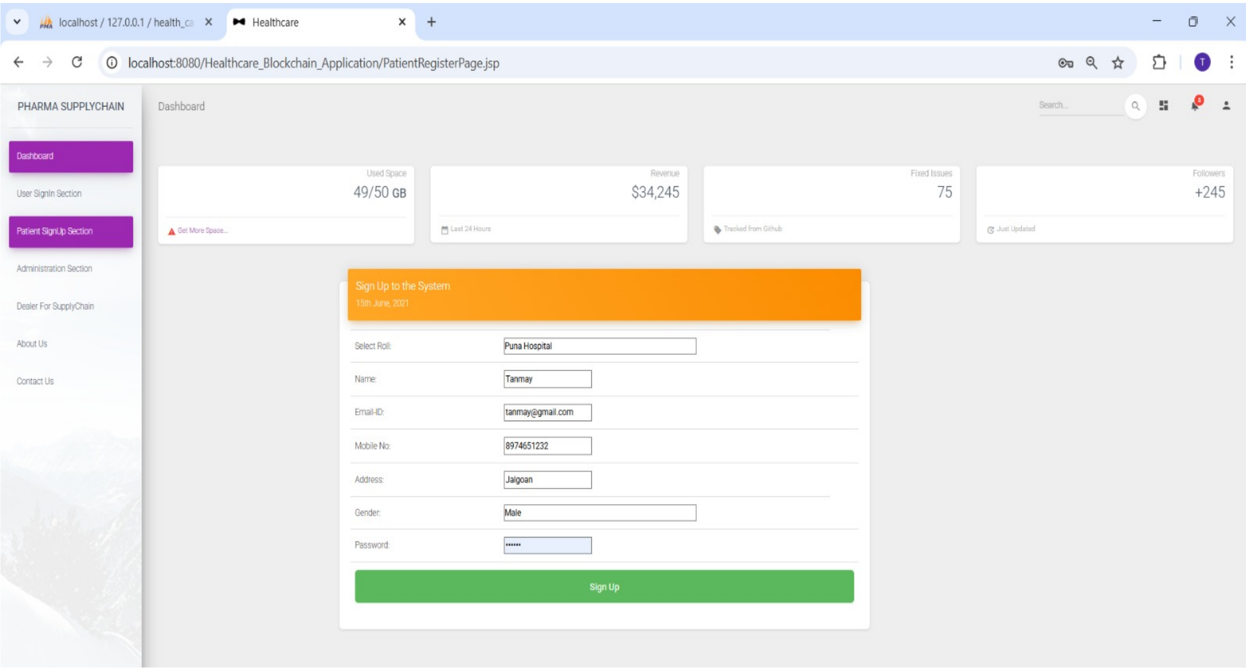

Figure 6.5: Running Project Snapshot 1

To manage pharmaceutical supply chain operations, maintain inventories, and keep an eye on shipments in real time, use your customized dashboard. Safe access for merchants, distributors, and manufacturers.

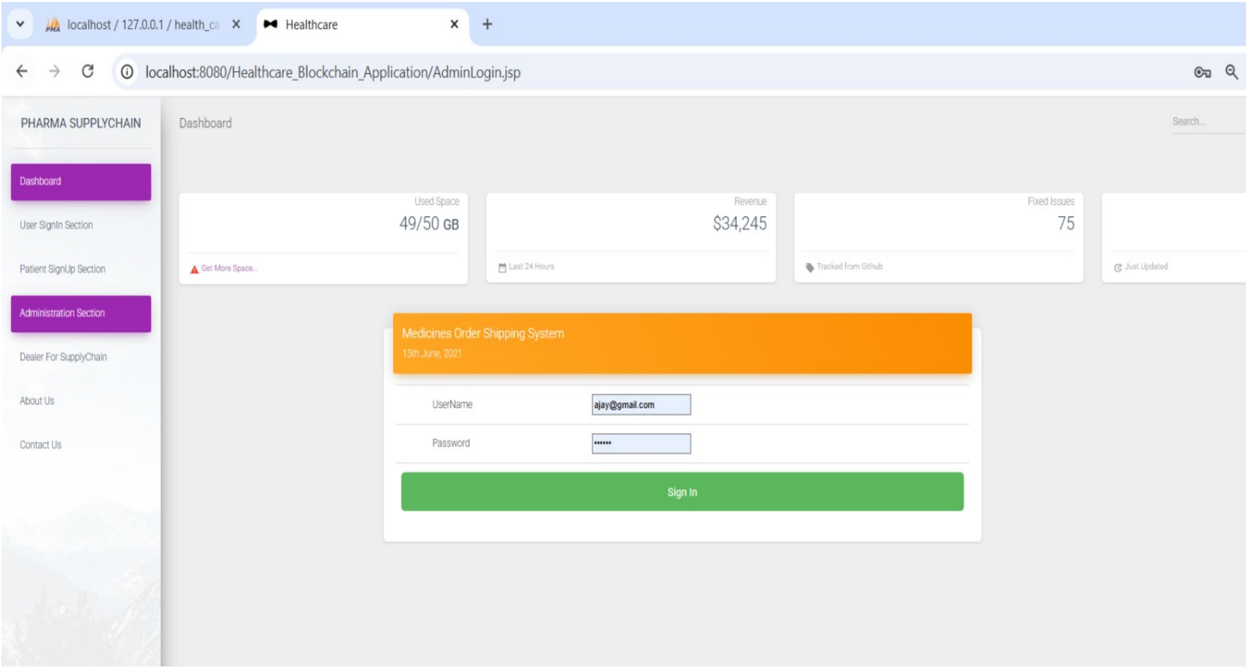

Figure 6.6: Running Project Snapshot 2

To manage pharmaceutical supply chain operations, maintain inventories, and keep an eye on shipments in real time,

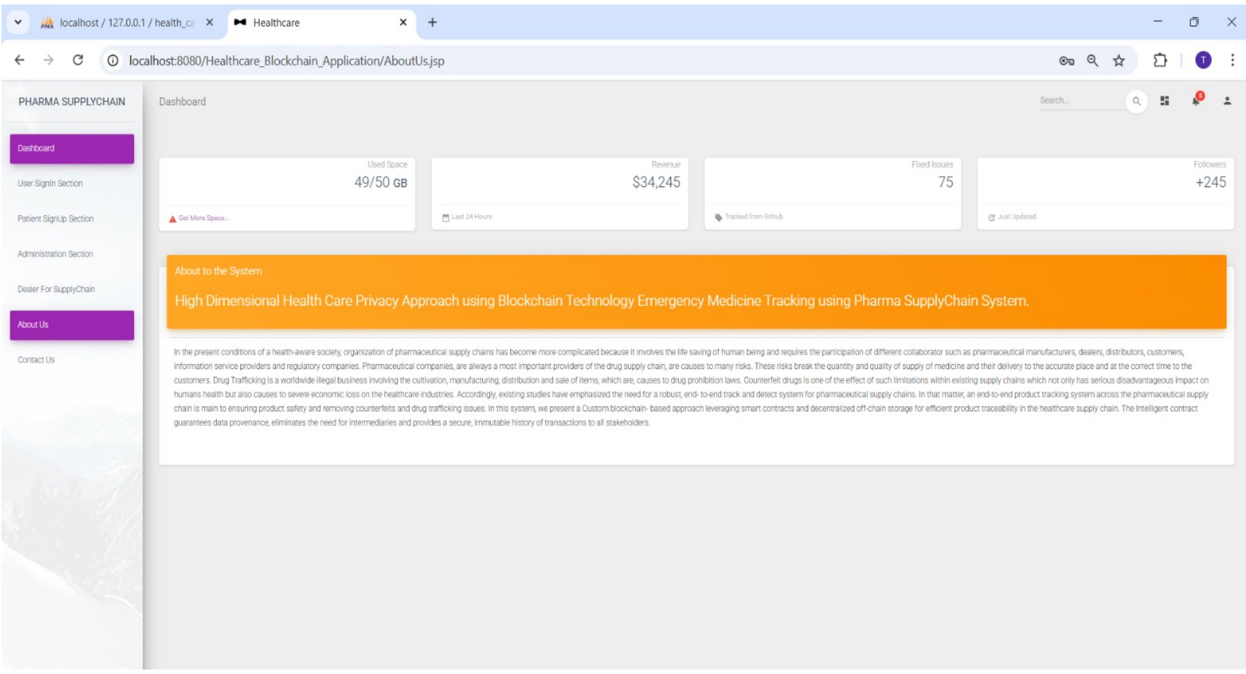

Figure 6.7: Running Project Snapshot 3

The user interface (UI) is designed to be intuitive and user-friendly, allowing patients, hospital staff, administrators.

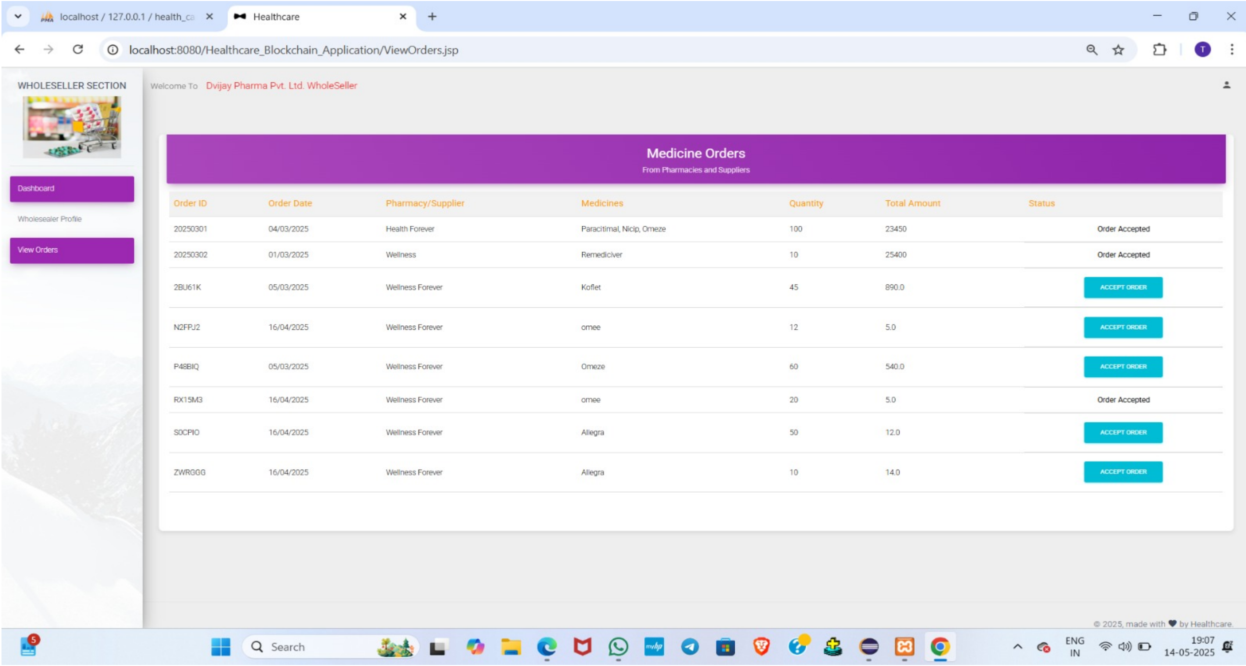

Figure 6.8: Running Project Snapshot 4

The user interface (UI) is designed to be show the user id ,password etc.

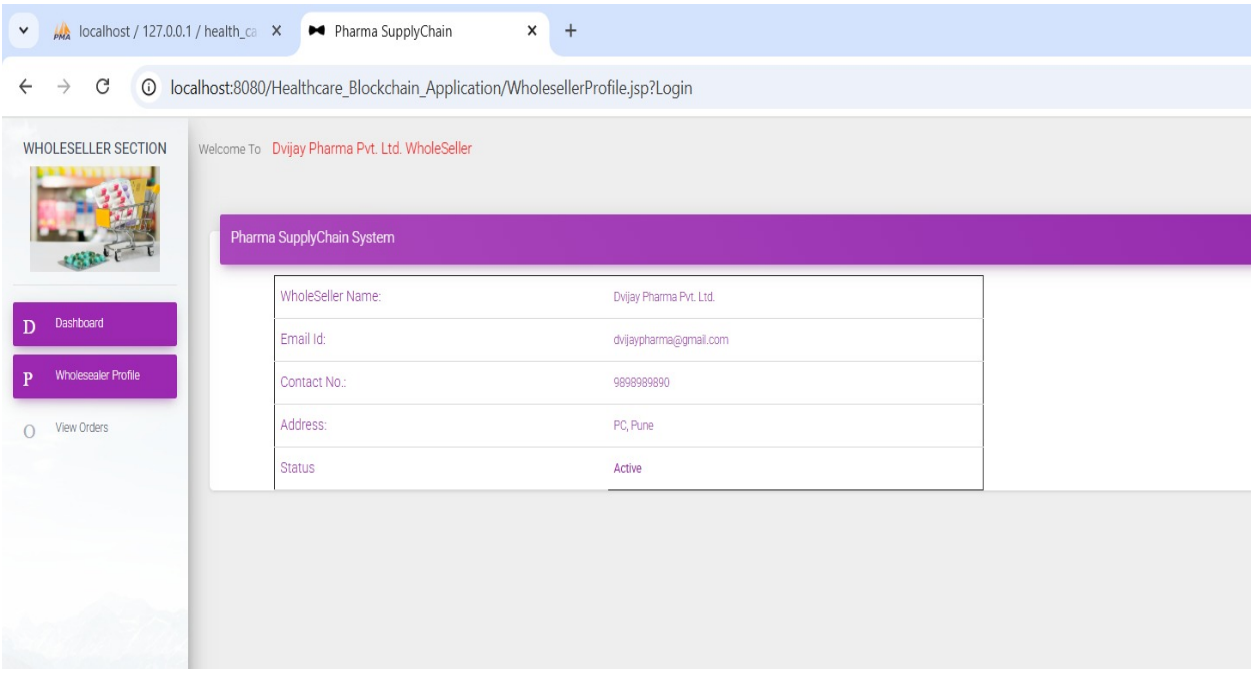

Figure 6.9: Running Project Snapshot 5

The user interface (UI) is designed to be intuitive and user-friendly, allowing patients, hospital staff, administrators.

6.2 Implementation of Modules

1. **User Interface Design** The user interface (UI) is designed to be intuitive and user-friendly, allowing patients, hospital staff, administrators, and pharma suppliers to navigate easily. It includes forms for data entry, dashboards for monitoring, and alerts for counterfeit risks, ensuring that all users can efficiently access the information they need.

2. **Web Project Creation in Eclipse** The web project is developed using Eclipse, where a structured environment is set up to manage different components of the application. This includes creating the project structure, defining modules, and integrating necessary libraries for blockchain functionality and web development.

3. **Database Creation** A relational database is created to store all necessary data, such as patient records, drug inventory, user details, and transaction logs. This database is designed with appropriate tables and relationships to ensure efficient data retrieval and storage.

4. **Database Connectivity** Database connectivity is established using JDBC (Java Database Connectivity) to facilitate communication between the web application and the database. This allows the application to execute SQL queries to retrieve, insert, update, and delete data as required by different modules.

5. **User Authentication** User authentication mechanisms are implemented to ensure that only authorized individuals can access the system. This includes secure login processes where users provide credentials (username and password) that are verified against stored data.

6. **User Verification & Validation** Upon login, user verification processes check roles (e.g., patient, admin, hospital staff) to grant appropriate access levels. Additionally, input validation is performed on all user inputs to prevent errors and security vulnerabilities, ensuring data integrity throughout the application.

These implementation steps work together to create a robust and secure system for preventing counterfeit drugs in the pharmaceutical supply chain using blockchain technology.

Chapter 7

Conclusions and Future Scope

7.1 Conclusion

The project demonstrates how blockchain technology and smart contracts can revolutionize the pharmaceutical supply chain in the healthcare sector. By implementing a novel Blockchain approach, the initiative addresses significant issues such supply chain transparency, regulatory compliance, and counterfeit medication. While the use of an immutable ledger ensures that every transaction is securely recorded and traceable, smart contracts automate and enforce compliance, reducing errors and fraud.

The system enhances the effectiveness and integrity of the pharmaceutical supply chain by providing a robust framework for verifying the authenticity of medications and safeguarding public health. This solution raises stakeholder trust and sets a new standard for accountability and transparency in the industry through secure data management and real-time tracking. By ensuring that patients receive genuine and effective medications, the endeavor ultimately makes the pharmaceutical business safer and more reliable.

7.2 Future Scope

By incorporating AI-powered anomaly detection, the Pharma Supply Chain-based Counterfeit Drug Prevention System may be further improved to more effectively detect counterfeit medications. Real-time tracking with IoT-enabled smart packaging is one potential future development that would enable stakeholders to keep an eye on drug authenticity at every turn. Furthermore, cross-border blockchain interoperability can guarantee smooth cooperation between various pharmaceutical supply chains around the world. Drug delivery may be made safer, more transparent, and more secure by strengthening the system further by adding decentralized AI-driven fraud detection models and extending smart contract automation to manage regulatory compliance.

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